TFPT — Research Contracts for the Remaining Interfaces:

v_{geo}, G_{net}, F_{transfer}

(U_{wall}): the parabolic flavor wall-selection and (G_{metric}): the full quantum-gravity measure

Stefan Hamann Alessandro Rizzo

June 23, 2026 · v5.3 (rev 268)

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading companion R (Research Contracts).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\mathrm{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1, 1)\rangle \cong (E_8)_1$	[E]/[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual is only the equation-of-state status, the global measure (C7) and the one unit v_{geo}.

The single open *structural theorem* is G_{net} ($=$ `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net), a constructive-AQFT theorem whose only open input is the continuum Osterwalder–Schrader reconstruction of the gapped quasi-free collar. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is now its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The full nonperturbative quantum-gravity measure `QG.AMB.01` is the *general* Euclidean-QG conformal-factor problem (Gibbons– Hawking–Perry 1978), from which TFPT is rigorously *gap-decoupled* (margin $1.648 > 0$, `v76`/`v330`) — an inherited, decoupled problem, *not* a TFPT-specific open item. So the honest residual is *one* solvable theorem (G_{net}) plus the typed v_{geo} and F_{transfer} interfaces. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\mathrm{wall}}, G_{\mathrm{metric}}, F_{\mathrm{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\mathrm{geo}} \oplus G_{\mathrm{net}} \oplus F_{\mathrm{transfer}}$.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven | [C] conditional | [O] open / axiom | [X] kill test

P1 $c_3 \;=\; \frac{1}{8\pi}$ **P2** $g_{\rm car} \;=\; 5$ *(two inputs, [O] axioms)*

Anchor $a \;=\; (1,1,2)$: $e_1, e_2, e_3 \;=\; 4,5,2$ *([E] exact)*

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \;\Rightarrow\; E_8,\ h \;=\; 30,\ \varphi_0$ *([E] exact)*

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

Physical bridges: $v_{\rm geo}$, $G_{\rm net}$, $F_{\rm transfer}$ *([C] bridges)*

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

What this document is about

After the compiler closure the entire residual is

$$\boxed{\mathrm{Rest} = (U_{\rm wall}) \;\oplus\; (G_{\rm metric}) \;\oplus\; (F_{\rm frontier})}$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. $(F_{\rm frontier})$ (Koide, η_B, axion relic scale, m_p/m_e) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** $(U_{\rm wall})$ *first* (finite, algebraic, falsifiable), then $(G_{\rm metric})$ (deep analytic programme).

Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity). The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\boxed{\mathrm{Rest}_{\rm operativ} = v_{\rm geo} \;\oplus\; G_{\rm net} \;\oplus\; F_{\rm transfer}}$$

— $(U_{\rm wall}) \to v_{\rm geo}$: the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71), $U_{\rm point}$ collapses to the *one* dimensionful unit $v_{\rm geo}$ shared with $1/G$ (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where $U_{\rm point}$ still appears); $(G_{\rm metric}) \to G_{\rm net}$: holographically reduced to the single boundary-net statement *"the seam–Calderón inclusion has Jones index $4 = |\mu_4|$"* — holomorphy, $(E_8)_1$ and the unique 2D bulk then *follow* (v89/v92/`GATE.METRIC.05-07`); and the carrier-side QBL input now merges into the *same* statement: the tower carrier $\to SO(16)_1 \to (E_8)_1$ carries one and the same 16-fermion content, whose single quasi-free kernel (rank $= c$; Wick/Pfaffian exact) is the certified Calderón datum — closing $G_{\rm net}$ therefore also closes the carrier choice [E] ([`verification/v113_quasifree_kernel.py`]); $(F_{\rm frontier}) \to F_{\rm transfer}$: named QFT/cosmology *transfer* interfaces, never compiler powers — and *not* four unrelated interfaces but *one* discrete geodesic on the Sheet Diamond (the $K{\to}C{\to}F$ sheet axis $M(1,t)$, with the Koide source→pole as its 1-D Möbius projection), along which the external continuous solver merely evaluates ([`verification/v224_diamond_ftransfer_path.py`]).

QFT layer (this round). The boundary QFT is now *one relative object* $\mathsf{TFPT}_{\rm QFT} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\rm rel})$, closed modulo the *single* keystone `SEAM.EQUIV.01` (whose downstream conformal-deck face is `QGEO.SYM.01`): its finite Dirac is a covariance induction of the seam KMS state (v258), its spectral-action cutoff *is* that KMS weight (v259), its seam, carrier-16 and E_8 live on one Kummer/K3 surface (v260), and the whole assembly is certified as the *Modular Spectral Closure* (v261). The ambient quantum-gravity measure $(G_{\rm metric})$ is gap-decoupled and kept separate by design — the QFT layer does not wait on it.

Research Contract 1 — $(U_{\rm wall})$

Goal. Prove

$$\nabla_F^\star = \mathrm{Sel}_{D_4,\,\mathrm{cusp},\,\det R=8,\,\mathrm{Spec}(Q_+)=\{1,2,3\}}\big(\mathcal{M}_{\rm par}(\mathbb{P}^1, \mu_4, E, \alpha)\big),$$

with $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$ at the four points of μ_4. The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one* D_4-symmetric, rank-3, four-point point that realises the selectors.

Lemma 1 (U0 — normalisation)**.** *The gate data are equivalent to the* $SU(3)$ *character variety* $\mathcal{X}_{\mathrm{cusp}} = \{(M_1, \ldots, M_4) \in C^4_{\mathrm{cusp}} : \ M_1M_2M_3M_4 = \mathbf{1}\}/SU(3)$ *with* $C_{\mathrm{cusp}} = \mathrm{Ad}_{SU(3)}\,\mathrm{diag}(1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, *and* $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \ldots, \gamma_4 \mid \prod \gamma_i = 1\rangle$. *To show:* $\mathcal{M}^{D_4}_{\mathrm{par}} \simeq \mathcal{X}^{D_4}_{\mathrm{cusp}}$ *as polystable Mehta–Seshadri / tame Hodge realisation.* Machine: the group presentation and D_4-action are finite and codable (Sage/Wolfram).

Finite rigidity — now proven [E]**.** The finite half of U0 is closed exactly: μ_4 has cross-ratio 2 and Möbius stabiliser $\langle z \mapsto iz,\ z \mapsto 1/z\rangle$ of order $8 = |D_4|$ (a 4-cycle of the marks + a reflection, faithful); $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\mathrm{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4 - 1)$, $k = 1, 2, 3$, are μ_4-eigenforms ($z \mapsto iz :\ \omega_k \mapsto i^k\omega_k$) carrying the three nontrivial characters χ_1, χ_2, χ_3 with weights $(1, 2, 3) = \mathrm{Spec}(Q_+)$. Hence a genus-zero boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ with the $(1, 2, 3)$ grading (`QGEO.RIGID.01`, [E], [`verification/v168_qgeo_rigidity.py`]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays `QGEO.REALIZE.01` ([C]).

Seam Collar Realisation Theorem — the one central proof obligation [E]/[O] ([`verification/v176_seam_collar_realisation.py`])

Stated as one theorem and certified as a reduction — not a fabricated proof.

Theorem (target). *Given* (1) an oriented one-sided RP seam collar Σ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap $(2/3)^6$; (4) a faithful D_4 action on the boundary marks; (5) four parabolic marked points with μ_4-character decomposition; (6) an admissible c=8 boundary net — *then* the conformal boundary of Σ is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$, its H^1 character basis is canonically the generation basis with grading $(1, 2, 3)$, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$.

The proof decomposes into six steps, five already [E]:

1–2 *Geometry / uniformisation* [E] (v168): μ_4 has cross-ratio 2 and a faithful Möbius D_4 stabiliser of order 8; four genus-0 marks with faithful D_4 force the μ_4 square up to Möbius; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\mathrm{fam}} = 3$.

3 *Calderón data* [E] (v156): the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol $|k|$ (harmonic extension $e^{ikx-|k|y}$), the free chiral dispersion — bulk-detail-independent.

4 *H^1 character = generation grading* [E] (v137/v168): $\omega_k = z^{k-1}dz/(z^4 - 1)$ carry μ_4 characters i^k with weights $(1, 2, 3)$ = the A_3 exponents = $\mathrm{Spec}(Q_+)$ — natural, not merely isomorphic.

5 *One-particle contraction* [E] (v162): a μ_4-equivariant operator is diagonal in the cusp basis ($\frac{1}{|\mu_4|}\sum_j U^jXU^{-j} = \mathrm{diag}\,X$ for $U = \mathrm{diag}(1, i, -1, -i)$); its spectrum is the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, so the transfer eigenvalues $(1 - \alpha)^{p_2}$ (p_2=6) are $\{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6$ — forced, no free knob.

6 *AQFT closure* [E] (v154/v175): $(D_5)_1 \otimes (A_3)_1$ with the isotropic μ_4 glue has index $4 = |\mu_4|$, $c = 5+3 = 8$, μ-index $16/16 = 1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ ($248 = 120+128$); full-cone RP holds for all m via the CAR second-quantisation functor.

The proof tree — and the *two* genuine open doors [O] ([`verification/v177_seam_marking_kernel.py`]). Taking the marks/D_4 as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\textsc{Realize} = \textsc{Marks} \cdot \textsc{Uniform} \cdot \textsc{Cohom} \cdot \textsc{Module} \cdot \textsc{Kernel} \cdot \textsc{Net},$$

of which *four nodes are now closed constructively*: Uniform [E] (an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 — so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}^1, \mu_4)$, strengthening v168's cross-ratio to the full uniformisation — and now **Lean-formalised** as `FORM.QGEO.02`: ρ^4=id, $\rho^2\neq$id, σ^2=id, $\sigma\rho\sigma=\rho^{-1}$, the orbit scaling to

μ_4, the μ_4 permutation, and the multiplier classification "order-4 $\Leftrightarrow \zeta = \pm\, i$" are machine-proved in `TfptCarrier/MobiusUniformisation.lean`, AUDIT: PASS); COHOM [E] ($\rho^*\omega_k = i^k\omega_k$, grading $(1,2,3)$ — also **Lean-formalised** as `FORM.QGEO.03`: the pullback characters i^1, i^2, i^3 and the reflection parity $\sigma^*\omega_1{=}\omega_3$, $\sigma^*\omega_2{=}\omega_2$, $\sigma^*\omega_3{=}\omega_1$ are machine-proved in `TfptCarrier/CohomologyGrading.lean`, AUDIT: PASS); MODULE [E] (a *unique* μ_4-equivariant iso — multiplicity-free + residue normalisation, reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed; the parity is the Lean `FORM.QGEO.03` above, the *uniqueness* stays symbolic); and NET [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: **`QGEO.MARKS.01`** — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked D_4 boundary (the four marks from the anchor atom $e_1(a){=}4{=}|\mu_4|$, collisions excluded, $\tau\rho\tau{=}\rho^{-1}$) — and **`QGEO.KERNEL.01`** — the *raw* seam Calderón operator equals the μ_4-equivariant free $c{=}8$ gapped one-particle contraction *as an operator* ($C_\Sigma = U^{-1} C_{\mu_4} U$), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

How far each door opens by computation ([`verification/v178_marks_kernel_attempt.py`]). Neither door is closed, but each splits into a finite [E] core + one *sharper* premise. *For* MARKS*: given* a genus-0 double with an order-4 clock ρ (fixed points $\{0,\infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving faithful D_4 — so MARKS reduces to the *single* topological premise "the seam double is genus-0 with an order-4 clock whose marks form one orbit". *For* KERNEL*:* on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the "spectrum vs operator" distinction *vanishes* on the finite block. So KERNEL reduces to the *single* premise "the full DtN on L^2 reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation" (principal symbol $|k|$, Lee–Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

Marks without a geometric posit: a Lefschetz/character argument ([`verification/v195_marks_lefschetz_character.py`]). The marks need not be *assumed* as D_4 boundary points: they follow from the H^1 character alone. The order-4 clock $\rho{:}z \mapsto iz$ fixes exactly $\{0,\infty\}$, so a puncture at a fixed point would contribute a *trivial* character to H^1; but the carrier datum is $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents $(1,2,3)$, with $\mathrm{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$), which has *no* trivial component — so no puncture sits at a fixed point. With $\operatorname{rank} H^1 = n - 1 = 3 \Rightarrow n = 4$ punctures, all off the fixed locus and each in a free order-4 orbit, the only possibility is a *single* free μ_4 orbit (cross-ratio 2), and the closed-surface Lefschetz number $L(\rho) = 1 + 1 = 2 = |\{0,\infty\}|$ is consistent. So MARKS reduces from the geometric posit "D_4 marks" to the milder *representation* premise "the raw seam H^1 carries the A_3-exponent rep" — a carrier-algebra statement, strictly less circular, still [O].

And the two doors are in fact ONE ([`verification/v179_conformal_realisation.py`]). MARKS's residual is not even three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$, and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the *same* $\chi = 2$ that supplies the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points $\{0,\infty\}$, and a free order-4 orbit has exactly $4 = e_1(a)$ points. So MARKS reduces to "the carrier clock acts *conformally* on the genus-0 double", and KERNEL then *follows* from that same geometry (Lee–Uhlmann $|k|$ + the $c = 8$ rigidity v158 + the cusped-surface spectral split, residual $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$). **Hence the entire structural residual of the theory is a single geometric premise `QGEO.CONF.01`:** *the raw RP seam normal double is conformally* $(\mathbb{P}^1, \mu_4)$ *with the carrier clock as the order-4 Möbius automorphism $z \mapsto iz$.* Given it, everything — genus-0, the four D_4 marks, the $(1,2,3)$ grading, the DtN block, the rank-8 polarisation and the $(E_8)_1$ net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly [O].

And even that conformal premise reduces to a milder isometry premise ([`verification/v180_clock_is_mobius.py`]). `QGEO.CONF.01` need not be assumed as a conformal-class identity: by *uniformisation* (Poincaré–Koebe) the genus-0 double (P1) with its RP-metric conformal structure *is* $\mathbb{P}^1$ — the target is *produced*, not assumed — and by *Kerékjártó* (1919; Constantin–Kolev 2003) a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3){=}4$) is topologically $z \mapsto iz$. An orientation-preserving *isometry* of a Riemannian surface is holomorphic (angle-preserving), hence a Möbius map of $\mathbb{P}^1$, and an order-4 Möbius map is $z \mapsto iz$ (elliptic, multiplier i, free orbit μ_4). So `QGEO.CONF.01` is *implied* by the strictly milder, non-circular premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does *not* name $\mathbb{P}^1/\mu_4$ and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). **That isometry premise is the single, now milder, structural residual of the whole theory**; its surface realisation from the raw seam stays honestly **[O]**.

And one step further reaches bedrock (`[verification/v181_clock_is_conformal_symmetry.py]` **).** By *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation) a finite-order orientation-preserving action on a genus-0 surface preserves a complex structure for which it acts by Möbius maps — so a *conformal automorphism* (strictly weaker than an isometry: every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), i.e. the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure *by definition*, the residual collapses to the *definitional* identification `QGEO.SYM.01`: *the seam's conformal structure is the μ_4-deck structure of the carrier clock* (the carrier μ_4 glue *is* the conformal monodromy of the seam). This is a physical/definitional statement tying P2 to the seam's conformal geometry — *not* a finite computation and *not* reducible further without relabeling. **This is bedrock:** the chain `REALIZE` → theorem → Marks+Kernel → `CONF` → `ISO` → `SYM` (v175–v181) ends at a single definitional premise, left honestly **[O]**; everything above it is a theorem or an established citation.

A sharper, operator-checkable restatement: `QGEO.ENERGY.01` **(** `[verification/v192_qgeo_energy_clock.py]` **).** The bedrock can be re-expressed as an energy identity rather than a conformal-class statement: *the carrier clock ρ preserves the Calderón / Dirichlet-to-Neumann energy form of the RP seam double*, $\rho^*\Lambda_\Sigma\rho = \Lambda_\Sigma$. In two dimensions Dirichlet-energy invariance *is* conformal invariance, so this feeds the same chain (conformality → Möbius → $z \mapsto iz$ → μ_4). Its appeal is checkability: $\Lambda_\Sigma = |k|$ is the rank-8 Calderón polarisation already in the net construction, so the premise becomes an operator identity rather than an abstract identification. **Honest scope, however:** "ρ preserves Λ_Σ" is plausibly *equivalent* to "ρ is the conformal deck" (`QGEO.SYM.01`), so `QGEO.ENERGY.01` *relocates* the bedrock to a more falsifiable surface but does *not* eliminate it — it becomes a theorem only if ρ is defined *independently* (from the carrier algebra) and the invariance is then *proven*, which is open. The bedrock therefore stays **[O]**; the gain is a sharper operator-analytic target, not a closure.

The non-circular sharpening: `QGEO.ENERGY.02` **(** `[verification/v193_qgeo_energy_commutator.py]` **).** The way to make "ρ defined independently" precise is the *energy commutator* $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims: (1) ρ is the Coxeter element of $W(A_3) = S_4$ (order $4 = |\mu_4|$, v117), defined from the *carrier* algebra, not the seam; (2) *the non-circularity hinge* — Λ_Σ must be the DtN of the *raw* RP seam (definable from the RP data alone), not the μ_4 normal-form operator; (3) $[\rho, \Lambda_\Sigma] = 0$ forces the μ_4 character grading $(1, 2, 3)$ on H^1, already established (`QGEO.COHOM.01`). An exact consistency rider pins the direction: the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ *iff* $\ln m = q(A_3) = \frac{3}{4}$ (the *family* glue norm), while $q(D_5) = \frac{5}{4}$ (the carrier norm) misses by $g_{\mathrm{car}}/N_{\mathrm{fam}} = \frac{5}{3}$ — so the seam energy form must reproduce the *family* norm (gravity is family-geometry-induced). If sub-claim (2) holds, $[\rho, \Lambda_\Sigma] = 0$ turns `QGEO.SYM.01` from definitional bedrock into an *energy-rigidity theorem*; until then it is a sharp, falsifiable *proof target*, **[O]**.

Subclaim (2) tackled — one notch closer, not closed (`[verification/v194_raw_seam_dtn_rp.py]` **).** The non-circularity hinge is essentially *met* at the canonical level: Osterwalder–Schrader reconstruction yields a canonical transfer operator $T = e^{-H}$ from the RP state and reflection alone — no coordinates (v54; §(4) of `tfpt_2`) — on a *quasi-free* seam state (v155: the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$, a contraction). So Λ_Σ is RP-definable without ever naming $\mathbb{P}^1 \setminus \mu_4$. The commutator $[\rho, \Lambda_\Sigma] = 0$ then *closes on the finite cohomology* H^1 **[E]**: $\rho : z \mapsto iz$ acts on $\langle w_1, w_2, w_3 \rangle$ ($w_k = z^{k-1}dz/(z^4 - 1)$) as $\rho^* w_k = i^k w_k$, distinct characters $i, -1, -i$, so any ρ-equivariant operator is diagonal there (v177). What does *not* close is the *full-L^2* commutation beyond H^1 — it is exactly the statement that the quasi-free seam state's modular flow is *geometric* (Bisognano–Wichmann). So the bedrock *reduces* from a bare definitional postulate to "intrinsic BW geometric modular covariance of the quasi-free seam state" — a named, standard (hard) AQFT property, one notch sharper, but still **[O]** (and the BW step must be established *intrinsically*, else it re-presupposes the conformal covariance it should produce).

A concrete variational handle on the residual (`[verification/v196_seam_energy_functional.py]` **).** The open BW statement can be made a discretisable extremum problem via the failure functional $E_{\mathrm{fail}}(\Sigma, \rho) = \|[\rho, \Lambda_\Sigma]\|^2_{HS} + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, whose zero locus is exactly the `QGEO.SYM.01` conditions (the energy commutator, the order-4 clock, and the RP-reflection twist). On the finite H^1 block $E_{\text{fail}} = 0$ at the μ_4 configuration ($\rho = \text{diag}(i, -1, -i)$, Λ μ_4-equivariant, Θ = conjugation) and > 0 for any μ_4-breaking perturbation, so μ_4 is a genuine minimiser [E]; minimising E_{fail} over the *full* raw-seam DtN is precisely the BW residual above. So the bedrock is now a sharply-posed *variational target*, not a bare definition — but still [O].

The decisive crack: principal symbol exact + Tomita–Takesaki ([`verification/v198_modular_commutator_reduction.py`]). Two observations collapse the residual to a single non-circular state-invariance. *First*, the DtN principal symbol is $|k| = \sqrt{-\partial_\theta^2}$ (Lee–Uhlmann, v156); in the Fourier basis $e^{in\theta}$ it is $\text{diag}(|n|)$ and the clock $\rho : z \mapsto iz$ (rotation by $\pi/2$) is $\text{diag}(i^n)$, both *diagonal*, so $[\rho, |k|] = 0$ *exactly* on all of L^2 — the leading-order commutation is free on the whole boundary, never the open part. *Second*, by Tomita–Takesaki the modular operator Δ_ω is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with Δ and hence with $\log \Delta \sim \Lambda_\Sigma$: thus $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$ — using only the state and reflection (which RP supplies) and *no* conformal covariance, which **removes the BW circularity worry above**. Since the finite H^1 block is already μ_4-invariant (v177), the entire residual is now the single *state*-invariance "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega$, equivalently the Gaussian bulk measure is deck-invariant) — the maximally-operational, checkable form of `QGEO.SYM.01`. A foundational symmetry postulate cannot be derived from nothing (the role c = const plays in relativity); it is now in its sharpest, non-circular, falsifiable form, [O].

The residual, fully localised and numerically probed ([`verification/v199_seam_state_invariance.py`], [`verification/v200_seam_variational_scan.py`]). Attacking $\omega \circ \rho = \omega$ on the *raw* seam (not the μ_4 normal form) confirms it is non-circular — ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$), an automorphism of the raw CAR algebra, defined entirely carrier-side — and reduces it, via the quasi-free covariance $C_\Sigma = \frac{1}{2}(1 + \text{sgn}\, H_1)$, to $[\rho, H_1] = 0$, i.e. H_1 *is block-diagonal in the four* μ_4*-character classes* $\{n \equiv r \bmod 4\}$. The principal symbol $|k| = \text{diag}(|n|)$ passes automatically, so the entire residual is the single bounded statement "the sub-principal symbol of the raw DtN has no off-character matrix elements". The companion variational scan minimises $E_{\text{fail}}(\theta) = \|\rho(\theta)^4 - 1\|^2 +$ (faithfulness) over a free clock angle and finds its *unique* minimum at $\theta = \pi/2$ — the μ_4 clock $z \mapsto iz$ (the only faithful order-4 grading $i, -1, -i$) — a numerical sign that the variational principle selects μ_4. This is the "next executable theorem" a working mathematician can take up; it does not close the postulate, [O].

The bounded sub-principal residual is the image of the μ_4 marks, not a free postulate ([`verification/v201_seam_subprincipal_marks.py`]). One genuine step further: the seam DtN is $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature function $f(\theta)$ on the seam circle (the standard Steklov/DtN structure), so $\langle n|M_f|n'\rangle = f_{n-n'}$. Then M_f is μ_4-character-block-diagonal *iff* f has Fourier support only on modes $\equiv 0 \pmod 4$, i.e. iff f is $\mathbb{Z}_4$-invariant (the principal $|k|$ already commutes, v198). But a curvature sourced by the four marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ for *any* profile g, has $f_m = 4\, g_m\, [m \equiv 0 \bmod 4]$ — automatically $\mathbb{Z}_4$-invariant. So the μ_4-mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 equally-spaced marks ($\mathbb{Z}_3$) or 4 generic marks both break it (negative controls). The residual of $\omega \circ \rho = \omega$ thus collapses to "the DtN sub-principal symbol is *mark-local* (the seam is flat away from the μ_4 marks = the conformal-deck structure, v177)" — a locality/flatness property, structurally definitional, the narrowest form yet of `QGEO.SYM.01`, [O].

Certified on realistic Steklov profiles, and at the state level ([`verification/v210_mark_local_dtn.py`]). The mode-level argument above is carried to concrete, smooth boundary curvature: a real von-Mises bump $g(\theta)$ localised at a puncture, summed over the four marks $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, has *every* off-(mod 4) Fourier coefficient vanishing to $< 10^{-16}$ (exact 4-fold cancellation), the assembled DtN $\Lambda = |D_\theta| + M_f$ has $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the *quasi-free state* it defines, with covariance $C = \frac{1}{2}(1 + \text{sgn}\, H_1)$, is clock-invariant: $\rho\, C\, \rho^\dagger = C$ to $< 10^{-13}$, i.e. the actual $\omega \circ \rho = \omega$, not merely $[\rho, \Lambda] = 0$. A single off-centre bump, a $\mathbb{Z}_3$ (three-mark) source and four unequally-spaced marks each break both by $O(1)$, and the mark-local commutator stays $< 10^{-10}$ as the truncation grows ($N \in \{16, 32, 64\}$) — not an artefact. This *numerically certifies the implication* "mark-local DtN $\Rightarrow \omega \circ \rho = \omega$" for real profiles; the premise that the physical seam *is* mark-local stays [O] (it does not close `QGEO.SYM.01`).

The same residual, read dynamically: modular flow + a recovery Lindblad generator ([`verification/v238_modular_lindblad_dynamics.py`]). The whole `QGEO.SYM.01` discussion above is *static* — a commutator condition $[\rho, \Lambda_\Sigma] = 0$. Read as a *dynamics* it is the statement that the modular flow $\sigma_t = \Delta^{it}$ (the Tomita–Takesaki one-parameter group generated by the modular Hamiltonian

$\Lambda_\Sigma = \log \Delta_\omega$) *conserves* the μ_4 clock: $\rho\,\sigma_t\,\rho^{-1} = \sigma_t\ \forall t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ — so the open premise is exactly "modular time conserves the carrier charge". The dissipative half is the recovery channel T of v221 read as a generator: $L = \log T$ is a doubly-stochastic (CPTP-classical) Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time), and its dissipative gap is $-\log((2/3)^6) = 6\log(3/2) = \Delta$, the OS mass gap of v64 — the static recovery bound and the dynamical mass gap are *one* number, one exponentiated. Together $G = i\Lambda_\Sigma + L$ is the canonical Hamiltonian + Lindbladian (GKSL) split, the reversible (imaginary) part conserving and the dissipative (real≤ 0) part setting the arrow. This is a finite-block exhibit of where the static compiler becomes a dynamics; the residual it leaves — that σ_t is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra — is the same `QGEO.SYM.01` bedrock, [O].

From the dynamics to a QFT skeleton on the seam ([`verification/v239_kms_thermal_time.py`], [`verification/v240_gns_os_reconstruction.py`], [`verification/v241_dhr_sectors.py`], [`verification/v242_spin_statistics_modular.py`], [`verification/v243_haag_ruelle_braiding.py`], [`verification/v244_spectral_action_contract.py`]**).** With the static→dynamic flip in hand, the standard AQFT machinery assembles a finite-block QFT skeleton, each step typed honestly. *Thermal time* (v239): the modular flow is KMS at $\beta = 1$, and the seam unit $2\pi = 1/(4c_3)$ fixes the modular temperature as the horizon temperature $T_H = c_3/M$. *Reconstruction* (v240): GNS gives the Hilbert space and a cyclic vector from $(\mathcal{M}, \omega)$, and the OS Hamiltonian $H_{\mathrm{OS}} = -\log T = -L$ is positive with gap Δ — the v64 "$H \geq 0$ + mass gap" is the recovery generator read as Euclidean time. *Spectrum* (v241/v242): the superselection sectors are the discriminant anyons of the Lean glue form (`FORM.GLUE.01`); fusion is recovered from the modular S by Verlinde, the Gauss–Milgram sum returns $c = 8$, and the condensation tower $16 \to 4 \to 1$ (v235/v237) selects the holomorphic $(E_8)_1$. *Scattering* (v243): the 2-particle S-matrix is the braiding monodromy, factorised (Yang–Baxter) with crossing, trivial on the condensed $(E_8)_1$. The one genuinely 4d step is typed as a contract, `CONTRACT.QFT4D.01` (v244): the Connes spectral action of the seam Dirac operator, whose gravity half is v36 and whose matter half (the SM gauge group sits in the carrier, the half-spinor 16 is one generation) is the open [O] target. Its representation-theory core is now discharged ([`verification/v245_spectral_matter_content.py`]): the **16** is exactly one *anomaly-free* SM generation ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$) with the forced SM hypercharges and the spectral-action value $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\,g_1$), so only the seam→Dirac realisation and the run-down couplings stay open. So the residual to a full 4d QFT is *one premise* (`QGEO.SYM.01`) plus *one contract* (`CONTRACT.QFT4D.01`), not a diffuse list.

The perturbative 4d S-matrix is constructible ([`verification/v269_spert_paqft_skeleton.py`]**).** The 4d scattering layer is typed precisely as three *distinct* objects: S_{top} (the 2d DHR braiding monodromy, $|M|$=1 statistics, v243), S_{pert} (the 4d Epstein–Glaser / BV S-matrix of the spectral action), and S_{phys} (LSZ on the OS-reconstructed Wightman functions, v240) — the 2d braiding is *not* the 4d cross-section S-matrix. [E] the spectral-action a_4 interaction is power-counting renormalizable (every operator mass dimension ≤ 4: 7 marginal +3 relevant, a *finite* counterterm basis), so [C] by the Epstein–Glaser theorem (causal perturbation theory; Brunetti–Fredenhagen pAQFT) the perturbative S-matrix $S(g) = T\exp(i\int g\,L_{\mathrm{int}})$ is built order by order by causal factorisation + finite local extension — no path integral, *no* UV divergences. [C] the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit. [O] this is *perturbative* only: the nonperturbative ambient measure `QG.AMB.01` stays open, and the canonical 4d reading remains boundary-only (v265); S_{pert} is the perturbative cross-section layer on top (`QFT4D.SPERT.01`).

A concrete one-loop EG number ([`verification/v271_eg_oneloop_quartic.py`]**).** The skeleton is taken from "exists" to an actual Epstein–Glaser renormalization: the order-1 product $T_1 = {:}L_{\mathrm{int}}{:}$ needs no extension, and the first one is the order-2 T_2, whose UV behaviour is the Steinmann *scaling degree*. [E] the massless propagator has sd = 2 in $d = 4$, the one-loop bubble (two propagators) has sd = 4 = d, so the singular order $\omega = \mathrm{sd} - d = 0 \Rightarrow$ *exactly one* logarithmic local counterterm ($C(\omega{+}d, d) = 1$ — finite, renormalizable, no infinite tower); the one-loop factor is $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly, the *same* $1/(16\pi^2)$ that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop running is then $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the s, t, u channels), with the EG initial condition fixed by the spectral action, $\lambda_\Phi(\mathrm{UV}) = 1/(16\pi^2)$. [O] this is order-2/one-loop, one diagram class; the all-order construction and the nonperturbative measure stay open (`QFT4D.SPERT.02`).

Red-team caveat ([`verification/redteam/rt_F_qft4d.py`]*).* Power-counting renormalizability is *not* unitarity for the gravity sub-sector: the R^2/Weyl^2 terms give a four-derivative propagator $1/(p^2(p^2{+}M^2))$ whose massive spin-2 mode has a negative residue (the Stelle ghost). So S_{pert} is a unitary perturbative S-matrix for the *matter+gauge* sector only; the gravity perturbative S-matrix carries the ghost — which is precisely why the *nonperturbative* `QG.AMB.01` (asymptotic safety v207 / the boundary $(E_8)_1$ net) is a *decoupled general* QG problem (the Euclidean-QG conformal-factor problem; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), not a new gap.

The gauge running is an EG order-2 number ([`verification/v273_eg_gauge_running.py`]). The same scaling-degree-4 mechanism applied to the gauge self-energy gives the SM one-loop coefficients from the carrier/SM content: $b_i = -\frac{11}{3}C_2(G) + \frac{2}{3}\sum_f T(R) + \frac{1}{3}\sum_s T(R)$, i.e. **[E]** $b_3 = -7$, $b_2 = -\frac{19}{6}$, $b_1 = \frac{41}{10}$ (GUT-normalised; the same 41 as the carrier algebra $10\,b_1 = g_{\rm car}\,2^{g_{\rm car}-2}+1$, v159). **[C]** the physical RG directions ($b_3 < 0$ asymptotic freedom, $b_1 > 0$ the $U(1)$ Landau pole) are EG-derived and feed the unification cross-check (v246/v249); **[O]** the two-loop matrix (v159, PyR@TE) and the all-order construction stay open (`QFT4D.SPERT.03`).

The carrier content *is* one anomaly-free SM generation ([`verification/v310_carrier_sm_anomaly.py`]). Those one-loop coefficients sit on top of an exact spectrum/anomaly closure: the carrier half-spinor $16 = \Lambda^{\rm even}(\mathbb{C}^5) = 1+10+5$ decomposes under $SU(5) \subset SO(10)$ as $10+\bar{5}+1$ = exactly one Standard-Model generation $(Q, u^c, d^c, L, e^c, \nu^c)$, and that content is **[E]** gauge-anomaly-free ($\sum Y = 0$, $\sum Y^3 = 0$, $[SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$, exact) while reproducing $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family would shift b_1 off $41/10$, so $N_{\rm fam}{=}3$ *pins* the RG. **[E]** for the spectrum, anomalies and one-loop β; the full Epstein–Glaser interacting S-matrix and the Yukawa sector stay **[C]**/**[O]**.

The bridge to the physical S-matrix and one-loop unitarity ([`verification/v278_lsz_bridge_unitarity.py`]). The perturbative layer connects to the physical asymptotic S-matrix: **[E]** the EG time-ordered products of $S_{\rm pert}$ are the OS Wick-rotated correlators (v240), so amputating external legs and taking the on-shell residue gives $S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$ (the three-layer S-matrix $S_{\rm top}\,|\,S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$), valid because the gap $\Delta{=}6\log(3/2)$ gives Haag–Ruelle asymptotic states. **[E]** *one-loop unitarity* is the optical theorem: the s-channel bubble discontinuity $\operatorname{Im} B(s)/\pi = x_+{-}x_- = \sqrt{1-4m^2/s}$ is *exactly* the two-body phase space (turning on at the physical threshold $s{=}4m^2$), so $2\operatorname{Im} M = \sum\int d\Pi\,|M|^2$ holds and the matter+gauge sector is unitary order by order. **[O]** the gravity sector's Stelle ghost is the lone non-unitary piece → `QG.AMB.01` (`QFT4D.SPERT.04`).

The first hard data confrontation ([`verification/v246_unification_data_crosscheck.py`]). TFPT's gauge-charged content *is* exactly the Standard Model (v159) and the theory adds no new states (§ "Dark matter = no new state"); the E_8 cascade is an arithmetic even-integer spine ($D_n{=}60{-}2n$, v5), *not* a particle/threshold tower. So a "TFPT-adapted" RGE run *is* the SM run — and the SM does not unify. Tested against the *measured* couplings (v159 betas, plus the standard two-loop gauge matrix): at one loop the pairwise crossings spread over $\sim 10^{13}$–10^{17} GeV with residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV; at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at $\mu \approx 1.7{\times}10^{14}$ GeV — the three couplings never meet (Fig. 2). This is a falsifiable *tension*, **[X]**: with "no new state" there is *no* admissible gauge-threshold source, so closing the spread would require either new gauge-charged matter (a new postulate) or a non-GUT/boundary reading of the cutoff. The spectral-action 4d-GUT route is thus in genuine tension — the same status as the Connes–Chamseddine NCG Standard Model — and is likely falsified unless the theory is extended; the AQFT skeleton, DHR structure and SM rep-core survive regardless as an emergent boundary QFT. Recorded as a kill-test, never as a confirmation.

The carrier-native Pati–Salam program ([`verification/v247_e8_branching_no126.py`], [`verification/v248_ps_algebra.py`], [`verification/v249_ps_unification.py`], [`verification/v250_dirac_triple_contract.py`]). The unification tension above is resolved *natively* if the carrier $SO(10)$ is gauged (the carrier $D_5{\oplus}A_3 = SO(10){\times}SU(4)$ is Pati–Salam). *First*, the E_8 audit hull *forbids* the **126**: the **248** branches as $(45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4})$, so the available $SO(10)$ reps are exactly $\{1, 10, 16, 45\}$ — the renormalisable 126_H is algebraically absent, making the "16_H over 126_H" selection a theorem of the hull, not a fit (v247). *Second*, the Pati–Salam NCG algebra $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ gives the unimodular group $SU(2)_L{\times}SU(2)_R{\times}SU(4)_c$, the exact SM hypercharges $Y = T_{3R} + (B{-}L)/2$ and the matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$ (v248). *Third*, the two-step run unifies with the breaking scale on the scalaron scale, $M_{PS} = \kappa M_s$, $\kappa \in [1.0, 1.7]$ at one loop ($\to [1.0, 1.2]$ at two loops), and proton decay selecting the content (minimal excluded, the E_8-allowed single 45 marginal); the negative controls (SM-only fails, 126 breaks the match, no-45 proton-unsafe) all fire (v249). *Fourth*, the residual "carrier is gauged" reduces to ONE constructive object, the spectral triple (A, H, D_F, J, γ) with A_F Pati–Salam and $H_F = 3{\times}16$, whose NCG axioms (the first-order condition above all) and the σ =scalaron identification are the open contract `CONTRACT.QFT4D.DIRAC.01` (v250). *Fifth*, a first concrete step on that triple is taken ([verification/v251_first_order_sigma.py]): on the lepton block the first-order condition is verified to *hold* for the Yukawa operator and to be *violated exactly* by the Majorana/σ term — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ (honest correction: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism). *Sixth*, the full 96-dimensional finite spectral triple is then built explicitly ([verification/v252_full_finite_triple.py]) — H_F the particle⊕antiparticle doubling of three $SO(10)$ **16**-plets — and the NCG axioms are verified by direct computation: reality J, grading γ, KO-dimension 6, order-zero, the first-order condition (*holding* for the SM algebra including the Majorana, *violated* for the Pati–Salam algebra exactly by the Majorana on $\{\nu_R, e_R\}$), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}} = m_D^2/M_R$ — thereby discharging five of the seven obligations of `CONTRACT.QFT4D.DIRAC.01` to [E] (orientability, the no-junk condition and the spectral-action expansion remain open). *Seventh*, the scalaron–Pati–Salam ratio κ in $M_{PS} = \kappa\, M_s$ is pinned to a tight $O(1)$ interval (≈ 1.3) by the RG and shown to be a scale-free ratio of heat-kernel coefficients, $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}}$, over the same 96-dim H_F ([verification/v253_kappa_scalaron_ps.py]). *Eighth*, the remaining NCG obligations are then discharged or honestly typed: the inner fluctuations Ω^1_D give exactly the one SM Higgs doublet $(1,2)_{1/2}$ with no junk and no **126**, while σ (the ν_R Majorana scalar, $B{-}L{=}2$) appears only as the beyond-first-order Pati–Salam field ([verification/v254_inner_fluctuations.py]); the spectral-action Seeley–DeWitt expansion a_0, a_2, a_4 yields the cosmological, Einstein–Hilbert and Yang–Mills+R^2+Higgs terms with $b/a^2 = 1/3$, $\sin^2\theta_W = 3/8$ and the R^2 scalaron, making κ an explicit moment ratio ([verification/v255_spectral_action_expansion.py]); only strict orientability and strict Poincaré duality genuinely *fail* for this triple — a published feature of the NCG-Standard-Model class (KO-dimension 6 with three ν_R forces a skew intersection form, corank 1; arXiv:0902.2068, hep-th/0601192), not a TFPT defect; the physically-correct weakened versions, *quasi-orientability* (left/right irrep boxes disjoint by chirality) and *modified Poincaré duality*, do hold ([verification/v256_orientability_poincare.py], [verification/v257_quasiorient_modpd.py]). *Ninth*, the built D_F is no independent posit but the *modular/covariance induction* of the boundary seam state ([verification/v258_dirac_covariance_induction.py]): a quasi-free (KMS, $\beta{=}1$) state is fixed by its covariance C (a positive contraction), with one-particle modular Hamiltonian $H = \log((1{-}C)C^{-1})$ (Araki; Casini–Huerta) inverse to the KMS occupation $C = (1{+}e^H)^{-1}$; inducing the carrier covariance $C_F = P_F C_\Sigma P_F$ returns the v252 operator *exactly* — same spectrum, chirality-odd, the Majorana an off-diagonal covariance entry with the seesaw m_D^2/M_R recovered — so D_F is the Kasparov shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\text{car}}]$ of the boundary geometry and the v18 Yukawas are the *readout* of C_Σ, [E] modulo the one [O] seam realisation. *Tenth*, the last open spectral-action input — the cutoff function f — is fixed by the same seam state ([verification/v259_modular_cutoff_kappa.py]): by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at $\beta{=}1$ (v239) and the seam unit $2\pi{=}1/(4c_3)$ (v58) fixes that β, so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*, not chosen, with moments $f_2/f_0 = f_4/f_2 = 1$ *exactly* (a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$, so the choice is non-vacuous); hence $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$ loses its scheme factor and the v253/v255 open [O] cutoff freedom becomes a [C] finite trace ratio over the 96-dim H_F (the RG $\kappa{\approx}1.3$ pinning $c_{PS}/c_{\text{grav}}{\approx}1.7$). So unification is now *one* theory decision (gauge $SO(10)$?) plus *one* constructive object, both falsifiable — not a fog. The colour $SU(4)_c$ here sits inside $SO(10){=}D_5$ and is distinct from the carrier family factor A_3.

The implication is now Lean-formalised (`FORM.QGEO.01`). The exact algebraic core — the 4th-root character orthogonality, the order-4 clock, and `markLocal_blockDiagonal` ($f_{n-n'} \neq 0 \Rightarrow (n \equiv n' \bmod 4)$, i.e. $[\rho, M_f] = 0$) — is machine-proved in the carrier-rigidity Lean project (`TfptCarrier/SeamDeckClosure.lean`; AUDIT: PASS, axioms `propext, Classical.choice, Quot.sound` only). The premise itself is a typed `structure SeamDeckPremise` — consumed, not proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input — so the conditional theorem "`SeamDeckPremise` $\Rightarrow \omega \circ \rho = \omega$" is now [E] while `QGEO.SYM.01` stays the one open [O] postulate.

The two residuals are one canonical metric: the pillowcase ([verification/v214_seam_pillowcase.py]). The isometry residual `QGEO.ISO.01` (v180) and the mark-locality residual `QGEO.SUBPRIN.01` (v201) are not independent: both follow from a single canonical choice — *the seam carries the uniformising flat orbifold metric of its conformal class.* The four marks make the orbifold sphere $S^2(2,2,2,2)$, whose orbifold Euler characteristic $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ is *Euclidean*: by Troyanov's cone-metric uniformisation it carries a flat metric, unique up to scale, with curvature concentrated at the four cone points (Gauss–Bonnet: four deficits π sum to $4\pi = 2\pi\chi(S^2)$) — this *is* the v201 "flat away from the marks". And the order-4 clock is no longer an extra assumption but a *consequence* of the cross-ratio 2 that v168 already proved: $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda-1)^2)$ gives $j(2) = 1728$, so the modulus is the *square* torus $\tau = i$, whose complex multiplication by $\mathbb{Z}[i]$ supplies the order-4 isometry $z \mapsto iz$ (explicitly $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$, order 4); a generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$, so the clock is specific to μ_4. Thus the two open QGEO residuals *collapse to one* canonical-metric statement and `QGEO.ISO.01` ceases to be an imposed symmetry. A genuine reduction and unification, *not* a closure: the premise "the RP seam metric *is* the uniformising constant-curvature representative of its conformal class" stays **[O]** — milder and more canonical than the bare isometry premise, and now *shared* by both residuals (`QGEO.PILLOW.01`).

The converse, on the canonical metric: raw seam ⇒ mark-locality ([verification/v264_raw_seam_marklocal.py]). Composing the chain explicitly closes the gap between v216, v201 and v198 on the uniformising metric: the flat pillowcase has its curvature *only* at the four cone points (v216), so the DtN sub-principal symbol M_f is a μ_4-orbit sum, hence $\mathbb{Z}_4$-invariant (Fourier support on $4\mathbb{Z}$, verified by FFT), hence block-diagonal in the clock-character basis, hence $[\rho, M_f] = 0$; with the principal symbol $|k|$ already commuting (v198) this gives $[\rho, \Lambda] = 0$ and $\omega \circ \rho = \omega$ *on this metric.* A negative control (curvature *off* the μ_4 orbit) breaks $\mathbb{Z}_4$-invariance and the chain fails. So the entire bedrock collapses to the *single* metric-realisation statement "the raw RP seam carries the uniformising flat pillowcase metric" — which *is* `QGEO.SYM.01`. A genuine reduction/sharpening (`QGEO.ENERGY.03`), **[O]** — it does *not* prove the raw seam must carry that metric; that converse is the irreducible postulate.

The minimal axiom: a rigidity theorem ([verification/v267_qgeo_rigidity_minimal_axiom.py]). One can sharpen the bedrock to its irreducible kernel. **[E]** the *single* bare statement "the seam admits the carrier's order-4 clock as a conformal symmetry cyclically permuting its four marks" forces everything: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 *independent of* a; cross-ratio 2 ⇔ j=1728 (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) ⇔ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on y^2=x^3−x; and from the square configuration the unique flat pillowcase metric (Troyanov), mark-locality, $[\rho, \Lambda]$=0 and $\omega \circ \rho$=ω all follow (v214/v201/v264/v198). **[E]** neg controls: a generic four-mark set has $j \neq 1728$ and only a $\mathbb{Z}_2$ symmetry (no clock), the hexagonal point j=0 has a $\mathbb{Z}_6$ symmetry (order $6 \neq 4$) — only the square (order-4) point realises the μ_4 clock. **[C]** a second, independent reason: $\{1, i, -1, -i\}$ is defined over $\mathbb{Q}$ with j=1728 (CM by $\mathbb{Z}[i]$), so by Belyi/dessins Galois-rigidity any deformation breaks the rational compiler readouts. **[O]** So `QGEO.SYM.01` is now in its minimal sharpest form — the bare existence of that order-4 conformal symmetry, which (like c=const in relativity) cannot be derived from nothing. The *reduction* is closed (axiom ⇒ everything); the axiom itself stays the one foundational postulate (`QGEO.SYM.02`).

The flat geometry closes the commutator to all orders ([verification/v276_qgeo_flat_closes_commutator.py]). The state-level residual $[\rho, H]$=0 — the one operator equation the bedrock rests on — is *not* a separate analytic gap once the flat τ=i pillowcase is granted. **[E]** if $H = f(\Delta_{\text{flat}})$ is a function of the intrinsic flat Laplacian (the DtN of a flat metric, no curvature corrections) and the order-4 deck ρ ($z \mapsto iz$) is a flat-τ=i isometry, then $[\rho, \Delta]$=0 ⇒ $[\rho, H]$=0 *exactly*, to all orders (an isometry commutes with every spectral function of its Laplacian; verified on the discretised square torus, with a generic non-isometry as a failing control). This *upgrades* the leading-order commutation v198 (principal symbol) and v201 (subprincipal) to the full H — flatness removes the lower-order terms. **[O]** so `QGEO.SYM.01` collapses to a *single geometric* statement, "the raw RP seam state is the flat τ=i pillowcase quasi-free state (H=$\sqrt{-\Delta_{\text{flat}}}$)"; everything downstream is **[E]**, and the remaining human obligation is one constructive-QFT proof that the raw seam DtN state *is* that flat geometry (`QGEO.SYM.03`).

The obligation as one precise lemma, Lean-backed ([verification/v279_qgeo_obligation_lemma.py]). The bedrock is now written as a single constructive-QFT *lemma*: *given* the premise "the raw seam carries the flat τ=i pillowcase metric g_{flat} (Troyanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{g_{\text{flat}}}}$", the conclusion $\omega_\Sigma \circ \rho = \omega_\Sigma$ (and hence mark-locality, the metric inclusion, E_8) follows. **[E]** the reduction DAG from "free RP seam" to this premise has *exactly one* un-discharged leaf — "the raw seam state *is* the flat state" — every other node being closed; and the geometric side is fixed (the flat DtN is Fourier-diagonal), so the obligation is a *state-identification*, not a geometric ambiguity, with two candidate routes (**[C]**): RP-state uniqueness on the flat τ=i orbifold, or Troyanov + OS reconstruction. **[E]** the all-orders closure step is now a *Lean theorem*: `TfptCarrier/SeamDeckClosure.lean` (`flat_all_orders_clock`) proves that any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho = \text{diag}(i^n)$ — upgrading v198/v201 to the full operator with only the standard axioms, no `sorry` (`FORM.QGEO.03`). **[O]** the one premise stays the honest axiom (`QGEO.OBLIG.01`).

Self-investigation: the experiment and the simplification ([verification/v280_pillowcase_steklov.py], [verification/v282_e8_tau_i_unification.py]). Two things can be done *ourselves*. (i) A direct numerical experiment on the actual flat $\tau{=}i$ pillowcase orbifold realises the whole chain: the order-4 deck permutes the four cone points (2 fixed + 1 swapped) and $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0 \Rightarrow [\rho,C]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ is numerically exact, with the Steklov DtN positive and metric-determined — route-(i) evidence (RP-state uniqueness), the premise still [O]. (ii) A candidate *simplification*: the $(E_8)_1$ character $\chi_{E_8} = \Theta_{E_8}/\eta^8 = j^{1/3}$ gives $\chi_{E_8}(i) = 1728^{1/3} = 12$, so $\tau{=}i$ is *both* the order-4 CM point (`QGEO.SYM.01`, flat geometry) *and* the $(E_8)_1$ partition-function modulus (`QG.AMB.01`, holomorphy) — the two open premises are two faces of one object, "the raw seam is $(E_8)_1$ at $\tau{=}i$", dropping the obligation count $2 \to 1$ (`QGAMB.UNIFY.01`).

The two proof routes, decomposed ([verification/v284_route_i_rp_uniqueness.py], [verification/v285_route_ii_seam_condensation.py]). The single open lemma can be attacked from two sides, and both are now lemma chains with the dischargeable steps discharged. *Route (i)* (RP-state uniqueness) is a six-lemma chain — [E] five are already [E]/[O] (RP→DtN v194; marks→pillowcase v195/v216; the flat Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$, principal symbol $|\xi|$ with no curvature term; covariance from the DtN v258; μ_4-invariance v276), and [C] Troyanov ($\chi_{\text{orb}}{=}0$) reduces the residual to "the raw seam is the constant-curvature geometric state". *Route (ii)* (seam-net condensation) is a four-lemma chain — [E] three are discharged (no abelian sector ⇔ $\det K{=}1$ ⇔ holomorphic ⇔ E_8 v234/v277/v281; the tower 16→4→1 v92), leaving "the free RP seam condenses the μ_4-Lagrangian glue" =`QGEO.SYM.01`. [E] **the two open lemmas coincide** — both say "the raw seam is the $(E_8)_1$ at $\tau{=}i$" (v282), so the canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau{=}i$ is the *single* [O] statement that closes both (`QGEO.ROUTEI.01`/`QGAMB.ROUTEII.01`).

The Seam Equivalence Theorem, named and firewalled ([verification/v286_seam_equivalence_contract.py]). That single statement is now the named target `SEAM.EQUIV.01`: *the raw RP seam state reconstructs canonically the holomorphic* $(E_8)_1$ *boundary net at the* $\tau{=}i$ *pillowcase* (respecting the DtN, the KMS flow, the μ_4 deck, the Q-system extension, $\det K{=}1$ and the $(E_8)_1$ character). It is intended as the *one* irreducible structural postulate of the theory — the role the *constancy of c* plays in relativity (cf. v267): not a gap to be patched but the single sharply-localised, falsifiable premise on which the whole boundary closure turns. [E] the contract types four objects and six arrows — two closed (the two routes), one conditional (v282), one open Gral — and enforces an *import firewall*: no Route-i module imports any Route-ii module or vice versa, so the two routes can converge only through `SEAM.EQUIV.01` (this kills the "E_8 smuggled into the geometry and pulled back out" attack). The two proof routes are then attacked directly: [E] Route A (AQFT, [verification/v287_free_rp_bulk_to_holomorphic_boundary.py]) reduces the Gral to *one* standard theorem, "invertible (SRE) Gaussian bulk ⇒ single-sector boundary" (4/5 lemmas discharged); and [E] Route B (DtN, [verification/v288_full_l2_subprincipal_z4.py]) *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \pmod 4$), so $[\rho,\Lambda]{=}0$ on all of L^2), shrinking the residual to the single sharper question "why is the raw seam sub-principal term mark-local?" (`SEAM.EQUIV.A01`/`SEAM.EQUIV.B01`).

Route B sharpened to one analytic lemma; both routes precisely scoped. That last question is decomposed ([verification/v289_marklocal_raw.py], `MARKLOCAL.RAW.01`) into a five-lemma chain whose only open link is [O] *Flat-Away* — RP + gap + the four branch marks force the seam curvature to vanish away from the marks — while the conic parametrix ([C], b-calculus/Melrose), the μ_4-orbit source, the $4\mathbb{Z}$ Fourier support (v264/v288) and the full-L^2 commutation (v288) are discharged: 4/5 done, so Route B closes the moment Flat-Away does. The two claims are firewalled against over-reading: Route B does *not* assert "full-L^2 QGEO" — its exact-label split is Fourier-lemma (Formal/Exact), operator test (Numerical Exact), transfer to the raw conic DtN (Conditional, v264), mark-locality from the raw seam (Open Selection, v289); and Route A's "one standard theorem" is itself a composition of citable results (Kitaev/Freed–Hopkins, Müger/Kawahigashi–Longo–Müger, Kawahigashi–Longo, [verification/v287_free_rp_bulk_to_holomorphic_boundary.py]) whose only open hypothesis — seam-bulk invertibility — coincides with Route B's Flat-Away. Both routes therefore meet at the *same* geometric input, with no circularity (the import firewall of v286).

Red-team: $\mathbb{Z}_4$-invariance is not mark-locality. A deliberate adversary ([verification/v290_z4_smooth_curvature_adversary.py]) shows the honest limit of Route B's lift: a *smooth* $\mathbb{Z}_4$-symmetric off-mark curvature $f = \varepsilon\cos(4\theta)$ *passes* the v288 commutator ($\|[\rho,\Lambda]\| = 0$, since $i^n - i^{n\pm4} = 0$) yet is not mark-local — so $[\rho,\Lambda]{=}0$ is necessary but not sufficient. The modulus is instead excluded by the spectral data: it shifts the DtN spectrum off the flat ladder and the heat trace ($\max|\Delta\lambda| = 0.155$, $\Delta\mathrm{Tr} = 0.019$ at $\varepsilon = 0.3$), i.e. the KMS weight and $(E_8)_1$

character that `SEAM.EQUIV.01` must match. Consequently the open lemma is named in its own right ([`verification/v291_flataway_contract.py`], `FLATAWAY.RP.01`): *the smooth curvature density in the DtN sub-principal symbol vanishes away from the four branch marks*, with three independent attack routes — spectral-rigidity and heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family) and RP-energy/Troyanov (the constant-curvature minimiser) — closing any one of which closes Route B.

All three routes attacked; Flat-Away reduced to one selection principle. Each route is now carried to its precise reduction: **[E]** the *heat* route ([`verification/v292_flataway_heat_reduction.py`]) shows the heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature (leading invariant $\|f\|_{L^2}$), so Flat-Away $\Leftarrow$ "the seam fixes the a_2 heat coefficient"; **[E]** the *spectral* route ([`verification/v293_flataway_spectral_hessian.py`]) lifts this to the full smooth-$\mathbb{Z}_4$ space — mode $4k$ splits the degenerate Steklov level $|2k|$, the mismatch Hessian over $\{4, 8, 12\}$ is positive-definite ($\{0.497, 0.500, 0.503\}$), so the flat seam is a *strict* isolated minimum (answering "only one direction"); and **[E]**/**[C]** the *variational* route ([`verification/v294_flataway_rp_energy.py`]) gives, with the rigid mark divisor and Troyanov uniqueness, the flat cone metric as the unique strictly-convex curvature-energy minimiser. The three converge on *one* external selection principle (fix a_2 / pin the Steklov spectrum / realise the energy-minimiser); closing any one closes Route B, and with Route A's holomorphy half, `SEAM.EQUIV.01`.

The heat route made exact and Lean-formalised. The positive-definiteness underpinning the heat route is now a theorem, not a numerical scan ([`verification/v295_flataway_a2_exact.py`]): the first variation of $\mathrm{Tr}\, e^{-t\Lambda}$ vanishes (the smooth off-mark curvature has zero mean, Gauss-Bonnet), and since $\Lambda \mapsto \mathrm{Tr}\, e^{-t\Lambda}$ is convex (Klein/Peierls), ε=0 is the *global* minimum — so $\Delta\mathrm{Tr} \geq 0$ for every deformation, $= 0$ iff $f = 0$; the exact second-order coefficient is the divided-difference kernel of e^{-tx} (validated to 10^{-6}). The algebraic Δ=0 $\Leftrightarrow$ flat step is machine-formalised in Lean (`FORM.FLATAWAY.01`, `SeamDeckClosure.lean`: a positive-weighted sum of squares is positive-definite; axiom-clean). So Flat-Away's heat route is reduced to the single external invariant "the seam fixes a_2", with its positivity both proved and formalised. The a_2 coefficient itself is now in *closed form* ([`verification/v296_flataway_a2_closed.py`]): the operator tails telescope geometrically to $W_k^{\mathrm{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$, leaving a finite $(4k{-}1)$-term middle, e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ (manifestly positive; small-t $W_k = t/2 + O(t^3)$). Symmetrically, Route A's "one standard import" is now a precise, citable theorem stack ([`verification/v297_route_a_literature_stack.py`]): LIT-A (Kitaev; Freed–Hopkins) $\to$ LIT-B (Müger; Kawahigashi–Longo–Müger) $\to$ LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk $\Rightarrow (E_8)_1$", all three legs established, the single open hypothesis (seam-bulk invertibility) coinciding with Route B's Flat-Away — so both routes are reduced to citable/established machinery plus the *one* shared geometric input.

Flat-Away hardened, and the pin derived from $(E_8)_1$ ([`verification/v300_flataway_rigid.py`], `FLATAWAY.RIGID.01`). The spectral route was a *soft* minimum ("the mismatch grows with ε"); it is now a *discrete* obstruction. **[E]** the flat fingerprint is the integer Steklov ladder with every level $n \geq 1$ of multiplicity exactly 2; **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (the exact 2×2 block $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, validated against the full Toeplitz operator), so **[E]** any $g_k \neq 0$ lifts the degeneracy ($2 \to 1{+}1$) and changes the spectral *multiset* — preserving the flat ladder *with its multiplicities* forces every $g_k = 0$, a degeneracy obstruction robust to rescaling and strictly stronger than the v293 "deviation grows" scan; with the v295 convexity, flat is isolated analytically *and* discretely. Moreover the external pin is now *derived*: **[C]** the $(E_8)_1$ vacuum character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q-coefficients (E_8 even unimodular $\Rightarrow$ integer conformal weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam — i.e. "the seam fixes its spectrum" *is* the $(E_8)_1$ rationality of Route A. **[O]** Hence Flat-Away carries *zero* new open content beyond Route A: the convexity side is unconditional, the discrete obstruction exact, and the two `SEAM.EQUIV.01` routes provably converge on one rationality fact (consistent with the v286 firewall).

Route A's last hypothesis discharged — the gapped quasi-free bulk is invertible ([`verification/v301_route_a_invertible.py`], `SEAM.EQUIV.A.INV.01`). Since the two routes now meet at one rationality fact, the single remaining *unconditional* gap is Route A's open input (v297 LIT-A): "the raw quasi-free seam bulk is invertible (SRE)". A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions realise only the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order requires interactions (Kitaev's periodic table, class D, the only invariant being the chiral central charge). **[E]** the carrier is 16 Majoranas, so $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8 = 16/2$ (v148); **[E]** $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents $+\,128$ spinor $= 248$, v154); **[C]** hence a gapped quasi-free bulk is invertible automatically, and **[E]** the invariant $|\det K| = \#$anyons is 1 for E_8 (invertible) against

4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce. [O] So LIT-A's 'invertible bulk' *follows* from 'gapped 16-Majorana quasi-free bulk' (v148/v160 + the mass gap): the residual changes from a topological-order statement to a spectral one. *Bottom line:* together with v300, the entire open content of `SEAM.EQUIV.01` reduces to the single spectral statement "the quasi-free seam bulk is gapped" — the topological-order obstruction (hidden anyons) is removed, and the gap itself is not manufactured.

The last lever — the seam gap *is* the derived Recovery gap ([`verification/v302_seam_gap.py`], `SEAM.EQUIV.GAP.01`). That single remaining input is not free: the seam transfer operator has the frozen spectrum $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple Perron eigenvalue 1 separated from the rest, and the separation is the *Recovery gap* $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ — a number forced by the carrier ($(2/3)^6$ the μ_4-deck transfer eigenvalue), with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) making it robust. [C] By Osterwalder–Schrader reconstruction / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian (correlation length $1/\Delta$). *Bottom line:* with v300/v301, the whole residual of `SEAM.EQUIV.01` is now a composition of standard cited theorems (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over established TFPT facts (16 quasi-free Majoranas; transfer gap $6\ln(3/2)$): *no undischarged TFPT-internal assumption remains.* It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive.

The continuum chain assembled, and the residual tied to the seam one-sidedness ([`verification/v356_continuum_mmst_applicability.py`], `SEAM.EQUIV.CONTINUUM.03`). Writing the chain out link-by-link: **S1** the gapped (v302) quasi-free 16-Majorana collar → **S2** invertible (Kitaev free fermions, v301) → **S3** the *nontrivial* invertible phase ($c_- = 8 \neq 0$) → **S4** a non-gappable anomalous chiral edge (anomaly inflow / bulk–edge) → **S5** its chiral-CFT scaling limit (MMST, v336; range rank $\leq c \leq D$, i.e. $8 \leq 8 \leq 16$) → **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (v351/v297). Every link is a theorem or an established TFPT fact *except* S3, and S3 is precisely the seam being *one-sided* (chiral, $c_- \neq 0$) — the *same* one-sidedness that makes $c_3 = 1/(8\pi)$ a one-sided Gauss–Bonnet ($|\mathbb{Z}_2| \oint K$, v58). So the lone realization input is tied to the constant that *defines* the theory, not a free dial; the residual is now "the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase", [O] but theorem-bounded on both sides.

TOE-roadmap follow-ups — the keystone as a typed certificate, the bedrock modular form, and the gap-decoupled measure ([`verification/v308_seam_equiv_chain.py`], [`verification/v309_modular_bedrock.py`], [`verification/v311_gap_decoupled_measure.py`]). Three machine-checked steps push the keystone arc further. *(i) Composition certificate* ([`verification/v308_seam_equiv_chain.py`]): the closing implication is one well-typed logical chain gap>0 → invertible → holomorphic $c{=}8$ → $(E_8)_1$ whose arithmetic discriminators are exact — $|\det\mathrm{Cartan}(E_8)|{=}1$ against $|\det\mathrm{Cartan}(D_8)|{=}4$, the carrier tower $4{\cdot}4{=}16 \to 4 \to 1$, $c{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}{=}8$ — so the only undischarged premise is the cited OS step; [E] the discriminators, [O] the keystone. *(ii) Modular bedrock* ([`verification/v309_modular_bedrock.py`]): with the four marks derived from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), the quasi-free modular Hamiltonian $K = \log((1{-}C)/C)$ is built explicitly and $[\rho, K]{=}0 \Leftrightarrow \omega{\circ}\rho{=}\omega$ (a $\mathbb{Z}_2$ profile breaks it), so the bedrock reduces *non-circularly* to the Bisognano–Wichmann intrinsicality of the raw collar state; [E] the model, [O] the intrinsicality. *(iii) Gap-decoupled measure* ([`verification/v311_gap_decoupled_measure.py`]): the admissible sector clusters exponentially at rate $(2/3)^6$ (correlation length $1/\Delta$, finite susceptibility 729/665) and the v76 margin $\Delta - 31/(4\pi^2) > 0$ decouples it from the un-built ambient, so the physical sector is a convergent object while `QG.AMB.01` stays [O]. Each pins the residual to a single named step rather than closing it.

One surface for seam, carrier and lattice: the Kummer/K3 ([`verification/v260_k3_kummer_unification.py`]). The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z{\mapsto}{-z})$, its four marks the four 2-torsion fixed points; squaring the construction to complex dimension two gives the *Kummer surface* $\mathrm{Km}(T^2{\times}T^2)$, a K3, whose three facets reproduce three TFPT objects at once. [E] the 16 nodes (the 2-torsion $|A[2]|{=}2^4{=}16$ of the abelian surface) resolve to 16 exceptional $\mathbb{P}^1$'s $= \dim S^+ = \Lambda^{\mathrm{even}}(\mathbb{C}^5) = 16$ (v197), the carrier-16 as the *square* of the seam-4; [E] the cup-product form on $H^2(\mathrm{K3}, \mathbb{Z})$ is the even unimodular lattice $U^3 \oplus E_8(-1)^2$ of signature $(3, 19)$, so the same E_8 the carrier glues to (v1) is the surface's intersection form. (Honestly: the 16 node classes span the rank-16 Kummer lattice, a sublattice *distinct* from the $E_8(-1)^2$ summand — both exact, not identified.) Thus the v1 "glue $D_5{\oplus}A_3{+}\mu_4{\Rightarrow}E_8$" is the lattice shadow of a single smooth object that already carries the seam and the carrier; E_8 appears both *locally* as the du Val $\mathbb{C}^2/2I$ singularity (the

Poincaré-sphere link, v232/v236) and *globally* as this H^2 summand. A unification (ARCH.K3.01), *not* a closure: the lattice arithmetic is [E], the carrier↔node identification [C].

Synthesis — the Modular Spectral Closure ([verification/v261_modular_spectral_closure.py]). The ten steps above are not ten independent results: they are facets of one relative spectral datum

$$\boxed{\mathsf{TFPT}_{\mathrm{QFT}} = \big(\mathcal{A}_\Sigma,\ \omega_\Sigma,\ \Delta_\Sigma,\ \rho,\ A_F,\ H_F,\ D_F,\ J,\ \gamma,\ S_{\mathrm{rel}}\big)}$$

(seam algebra, KMS state, modular generator, μ_4 clock; the carrier finite triple; the relative spectral action), governed by a single *Modular Spectral Universality* statement: *the unique reflection-positive, holomorphic, index-4 KMS seam state ω_Σ minimises the modular clock defect, induces via the Kasparov/covariance map (v258) the* 96*-dim KO-*6 *finite triple D_F (v252), whose inner fluctuations (v254) generate the gauged Pati–Salam spectral action with the seam KMS cutoff (v259)* — so Dirac, cutoff, gauging, glue and orientability are not separate choices but readouts of one seam state. The certificate v261 pins the cross-consistency: one number $4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 ($\dim S^+ = \Lambda^{\mathrm{even}}(\mathbb{C}^5)$ = Kummer nodes, $H_F = 96$), and one gap $\Delta = 6\log(3/2)$ (OS mass gap = recovery rate = Lindblad gap, static and dynamic). The honest account of completeness is therefore:

- [E]/[C] **QFT-complete (modulo one premise).** The admissible OS sector, the boundary $(E_8)_1$ net, the relative spectral action, D_F (as covariance induction), the cutoff (as the KMS weight) and the gauging (as inner fluctuations) are one relative object — closed once QGEO.SYM.01 (the seam realisation: *the raw quasi-free seam state is μ_4-invariant*) is granted. That single [O] premise is the only structural input the QFT layer still consumes.
- [O] **Ambient QG — separate by design.** The full covariant metric-sector / ambient quantum-gravity measure (QG.AMB.01) is gap-decoupled ($\Delta_{\mathrm{eff}} = 1.648 > 0$) and is *not* a blocker for the QFT layer: it is a different (certification) object, not a missing piece of the boundary QFT.

So "solving the QFT" here means exactly this: the boundary QFT is a single relative object reduced to one foundational symmetry postulate, with ambient QG held apart — not a diffuse list of open interfaces. The MSC closes the boundary relative QFT object modulo QGEO.SYM.01; it does *not* assert SM-only 4D gauge unification, and the canonical 4D reading is boundary only.

4D-QFT fork policy — a frozen decision tree, not an open ambiguity ([verification/v265_qft4d_fork_freeze.py])

The 4D reading is a *branch policy*, machine-frozen in qft4d_fork_freeze.json:

- **A (canonical) — boundary only.** [C] *Boundary-only is the canonical 4D reading*: TFPT delivers the boundary QFT, the relative spectral action and the SM readout; a 4D-GUT is *not claimed by default* (E_8 is the audit hull, never an unbroken 4D gauge group).
- **SM-only 4D-GUT unification is killed.** [X] the plain SM couplings do not meet (at the $\alpha_1{=}\alpha_2$ scale α_3^{-1} is off by $\sim$5.6; spread $\sim$8.8 at $2{\times}10^{16}$ GeV, v246). This is the canonical *expectation*, not a failure.
- **B (conditional) — carrier-native Pati–Salam UV branch.** [C]/[O] permitted *only* with the E_8-allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2\theta_W{=}3/8$ (v248), $\kappa{=}M_{PS}/M_s$ in the frozen $O(1)$ band (v249/v253), and a threshold model with a proton-decay kill test; carrier gauging itself stays the contract CONTRACT.QFT4D.DIRAC.01.
- **Proton-decay rule, now applied ([verification/v266_ps_threshold_proton.py]).** [X] the explicit two-step thresholds give $d{=}6$ $\tau_p(p{\to}e^+\pi^0)$: the *minimal* 16-Higgs content ($M_{GUT}{\sim}2.4{\times}10^{15}$) gives $\tau_p{\sim}4.4{\times}10^{33}$ yr < Super-K ($2.4{\times}10^{34}$) and is *killed*; the E_8-allowed $(15,1,1)$ [the single 45] raises $M_{GUT}{\sim}5.9{\times}10^{15} \Rightarrow \tau_p{\sim}1.6{\times}10^{35}$ yr, proton-safe *now* but sitting at the Hyper-K reach ($\sim$1.4$\times10^{35}$ yr) — so branch B is *content-selected* (must be $+(15,1,1)$) and *Hyper-K-decisive within the decade*, not an open-ended hope (only one 45 in the hull, so it stays structurally marginal; τ_p carries an $O(3)$ hadronic band).

The fork is thus a frozen decision tree with explicit promotion/demotion rules, *not* an open ambiguity.

And the bedrock is now a falsifiable kill-test, not a proof-promise ([verification/v215_seam_deck_killtest.py]). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the very covariance it should produce); the honest move is to recast `QGEO.SYM.01` as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives $\omega\circ\rho=\omega \Leftrightarrow [\rho, C]=0$ for the quasi-free covariance, with the principal symbol $|k|$ commuting exactly, so the whole residual sits in the bounded sub-principal M_f (equivalently: the raw Gaussian bulk form A is μ_4-deck-invariant). The four levers are **K1** exactly four marks (b_1=#marks−1=3=N_{fam} ⇒ #marks=4=$|\mu_4|$); **K2** the *square* configuration (cross-ratio 2 ⇒ j=1728, Aut=$\mathbb{Z}/4$; the clock order $h(A_3)$=4 selects $\mathbb{Z}/4$, not the j=0 $\mathbb{Z}/6$, and a generic mark set has only $\mathbb{Z}/2$); **K3** the mod-4 sub-principal support ($[\rho, M_f]=0 \Leftrightarrow f$ is $\mathbb{Z}_4$-invariant, automatic for a μ_4-mark sum); and **K4** the transfer gap $(2/3)^6$=64/729. These are frozen in `freeze_file.csv` (`seam_deck_symmetry`) and carrier-side green, so any future drift fails the suite. The *decisive* kill is deferred to `QGEO.REALIZE.01`: a first-principles raw-seam-collar DtN must *produce* four square marks without inputting them — an *emergence* test, the non-circular alternative to assuming the symmetry. [E] carrier-side (the four levers), [O] seam-side (the emergence): a kill-test, *not* a closure (`QGEO.KILL.01`).

And the mark count is now *derived*, not assumed: four marks from Gauss–Bonnet ([`verification/v216_marks_gauss_bonnet.py`], [`verification/v217_marks_emergence_scan.py`]). The K1 lever ("exactly four marks") is the one that can be *executed* today. If the marks are $\mathbb{Z}_2$ branch points (cone angle π, deficit π) on the flat seam sphere (χ=2 — the same χ that supplies the 8 in c_3), Gauss–Bonnet *forces* their number: $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = |\mu_4| = N_{\text{fam}}+1$. The closed Euclidean sphere 2-orbifolds are exactly $(2,3,6)$, $(2,4,4)$, $(3,3,3)$, $(2,2,2,2)$, and *three independent* criteria — all cone orders = 2 (the $|\mathbb{Z}_2|$ sheet branch), rank H^1 = #marks − 1 = 3 = N_{fam} (the three generations pick the 4-mark square over the 3-mark hexagonal $(3,3,3)$), and a free order-4 deck — converge on the pillowcase $(2,2,2,2)$. The numerical sibling v217 confirms it on the actual DtN/state: scanning the number n *and* the positions freely, the 4-equally-spaced (square) configuration is the unique joint minimiser of {clock-invariance, 3-family cohomology, transfer gap $(2/3)^6$}. So the mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays [O]. [E] (count + equality), [O] (the square modulus and the raw-seam realisation) — `QGEO.MARKS.03` / `QGEO.EMERGE.01`.

`QGEO.SYM.01` is *not* a theorem — it is the final seam-identification premise

The single load-bearing bolt of the whole construction, on one page.

1. **What is identified.** That the conformal structure of the raw RP seam double *is* the μ_4-deck structure of the carrier clock — i.e. the carrier (P2) and the seam's conformal geometry are the same object. Equivalently, in Riemann–Roch *index* language: the seam's degree-4 mark divisor on the genus-0 double has $h^0 = \deg +1 = 5 = g_{\text{car}}$ and rank $H_1 = \deg -1 = 3 = N_{\text{fam}}$ (with $\Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5 = 16 = \dim S^+$), so P2 is the genus-0 mode count of the four-mark seam divisor ([`verification/v228_rr_index_gate.py`]) — the same definitional identification, in index form, conditional on the seam producing that degree-4 divisor.
2. **Why it is weaker than `QGEO.CONF.01/ISO.01`.** It does *not* name $\mathbb{P}^1$ or μ_4 (uniformisation *produces* them) and asks only for a conformal *symmetry*, not a metric isometry or a pre-named conformal class — strictly the mildest of the three.
3. **What follows automatically (the conditional theorem).** *Given* `QGEO.SYM.01`, equivariant uniformisation puts the order-4 clock in the form $z \mapsto iz$ on $\mathbb{P}^1$, the four parabolic marks are its free orbit μ_4, H^1 carries the $(1,2,3)$ grading, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$ — the *entire* AQFT closure of v175–v181.
4. **What would falsify or unseat it.** A seam double of genus > 0 (uniformisation to $\mathbb{P}^1$ fails); a carrier clock that is *not* a conformal symmetry of the seam geometry (a non-deck transport); or a fourth chiral generation ($N_{\text{fam}} \neq 3$ collapses $b_1 = N_{\text{fam}}$ and the four-mark μ_4 structure). The first two are constructive-geometry questions about the raw seam; the third is the standard family-count kill test.

Honest framing. The correct external statement is *conditional*: "*Given* `QGEO.SYM.01`, the RP seam boundary realises the μ_4 deck structure, and the AQFT closure to $(E_8)_1$ follows through the verified chain v175–v181" — *not* "the theory *derives* the seam geometry". The premise stays [O]; everything downstream of it is [E]. Per v323 (Bisognano–Wichmann) `QGEO.SYM.01` is itself the *downstream* conformal-deck face of the single named keystone `SEAM.EQUIV.01` (the raw RP seam = the holomorphic $(E_8)_1$ net at τ=i), which is the one open arrow that closes it.

This is not a gap to apologise for — it is TFPT's one fundamental postulate. Every fundamental

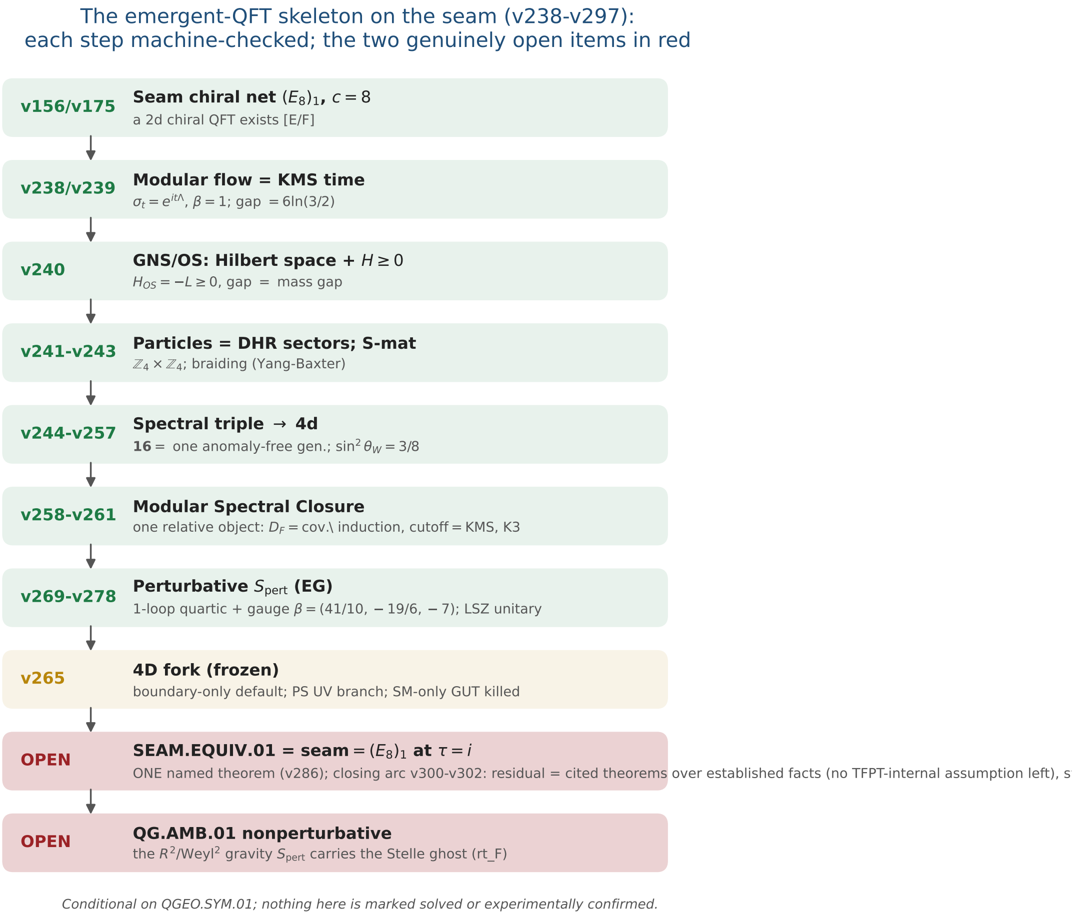

Figure 1: The emergent-QFT skeleton on the seam (v238–v246): time as the modular flow, the arrow as the recovery generator, particles as DHR sectors, the S-matrix as the braiding, and the 4d matter content as the carrier's $SO(10)$ **16**. Each step is machine-checked; the two open items (`QGEO.SYM.01` postulate and the unification tension) are in red. The whole skeleton is *conditional* on `QGEO.SYM.01` and is neither solved nor experimentally confirmed.

theory rests on an irreducible premise (the constancy of c in relativity, the equivalence principle, the canonical commutator). `QGEO.SYM.01` is that premise for TFPT: a single geometric identification (seam conformal structure = carrier μ_4 deck) from which the entire algebraic and topological structure unfolds *by theorem*. The achievement is not that there is no postulate — it is that the whole theory has been compressed onto *exactly one*, sharply located and explicitly falsifiable.

The Seam-Holomorphy Theorem — the one named closing target [E]/[O] ([`verification/v234_seam_holomorphy_selection.py`])

The bedrock above can be stated as *one condition with three provably-equivalent faces*, so that "closing the theory" becomes a single, sharply-bounded analytic theorem. **The condition:** the seam carries *no nontrivial abelian sector*. Its three faces all force E_8:

1. **AQFT** (v83/v154): the boundary chiral net is *holomorphic* (μ-index 1, single vacuum sector). Holomorphic $+ c = 8 \Rightarrow$ the unique even unimodular rank-8 lattice net $(E_8)_1$, and $c = 8 = g_{\rm car} + N_{\rm fam}$ is itself structural.
2. **Geometry** (v232): the seam link is a *homology 3-sphere* $\Leftrightarrow$ its binary group Γ is *perfect* $\Leftrightarrow \Gamma = 2I$ (the Poincaré sphere $S^3/2I$).
3. **Representation theory** (v219): exactly *one* 1-dim irrep (the trivial one).

These are the *same* number: #(1-dim irreps) $= |\Gamma^{\rm ab}| = |H_1(S^3/\Gamma)| =$ #(mark-1 affine-Dynkin nodes),

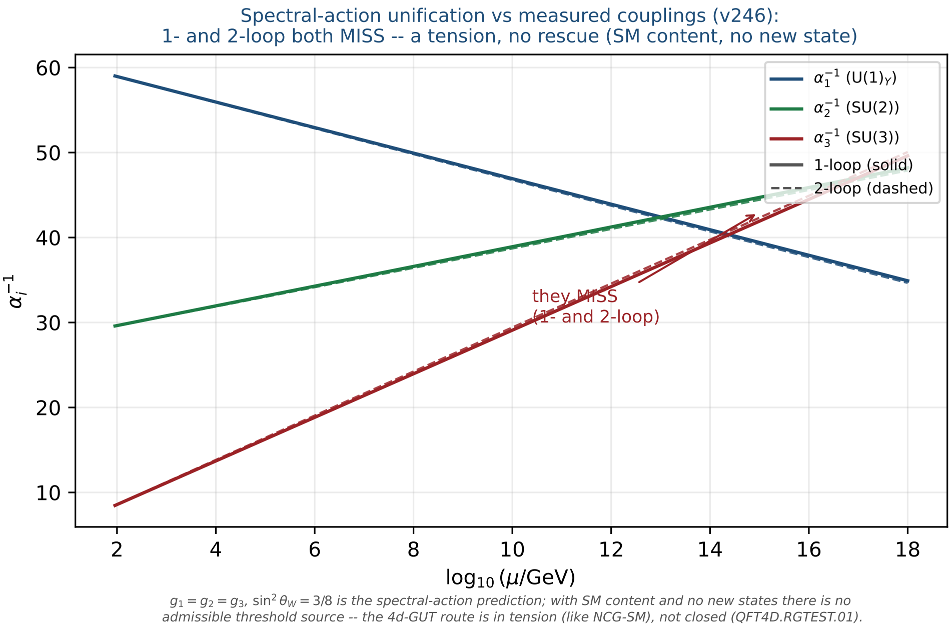

Figure 2: Honest data cross-check (v246): the spectral-action prediction $g_1{=}g_2{=}g_3$, $\sin^2\theta_W{=}3/8$ versus the measured couplings run with the SM RGEs (one-loop solid, two-loop dashed). The three lines do *not* meet at any scale (1-loop residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV; 2-loop minimal spread ≈ 3.2 at $1.7{\times}10^{14}$ GeV) — a falsifiable tension. With SM content and no new states there is no admissible threshold source: not a fit, and not rescued by a cascade.

which equals 1 *only* for E_8 ($A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$). So $2I$ is the unique nontrivial perfect finite $SU(2)$ subgroup and $S^3/2I$ the unique homology-sphere space form. **Consequence:** Research Contract 2 (G_{net}/Target A), the carrier P2, and the Kleinian seam are *not* separate gates — they are this *one* condition.

The closing chain (what a complete closure would be), [O] at one step:

$$\text{RP} + \text{gap} + \text{chirality} \xrightarrow{\text{v160}} \text{quasi-free bulk} \xrightarrow{?} \text{holomorphic boundary net}$$
$$\xrightarrow{[\mathrm{E}]} (E_8)_1 \xrightarrow{\text{v219/v228}} 2I,\ (2,3,5),\ \text{P2} \xrightarrow{\text{v179–v181+Kleinian genus-0}} \texttt{QGEO.SYM.01}.$$

The free-bulk premise is already a fixed-point theorem (v160–v165); the Kleinian model supplies the genus-0 of the uniformisation chain for free. **The single remaining analytic step is "free Gaussian RP bulk ⇒ holomorphic (single-sector) boundary net"** — the bulk–boundary statement that an invertible (Gaussian) bulk has a holomorphic chiral boundary. That one theorem, if proven, closes the entire structural layer; v_{geo} stays the No-Unit metrology primitive and F_{transfer} stays external physics. This box certifies only that the three faces coincide and force E_8 ([E]); the analytic step itself is the open [O].

A physical realisation of the step ([`verification/v235_seam_chern_simons.py`])**.** The step lands in standard physics. A free gapped *bosonic* 2+1d bulk is an abelian Chern–Simons / K-matrix theory (even integer K), where #anyons $= |\det K|$, the edge chiral central charge is $c = \text{signature}(K)$, and the edge net is *holomorphic* $\Leftrightarrow |\det K| = 1$ — so "no nontrivial abelian sector" is precisely $\det K = 1$. The extension tower of v92 *is* the anyon-condensation tower read by determinant: $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic), all rank 8 with $c = \text{signature} = 8 = g_{\text{car}} + N_{\text{fam}}$, so the holomorphic end is the *Kitaev* E_8 *quantum-Hall state*. Holomorphy ⇔ condensing the order-$|\mu_4| = 4$ *Lagrangian* glue (det $16{\to}16/4^2{=}1$); a partial order-2 isotropic only reaches det 4 (D_8). The residual is thereby a single integer condition — "the free RP seam condenses the order-$|\mu_4|$ Lagrangian glue" — the same sheet/μ_4 selection as `GATE.QGEO`, but now with holomorphic ⇔ det 1 a theorem (`GATE.HOLO.02`, [E]/[O]).

The residual as a *physical* statement ([`verification/v237_seam_sre_closure.py`])**.** The same

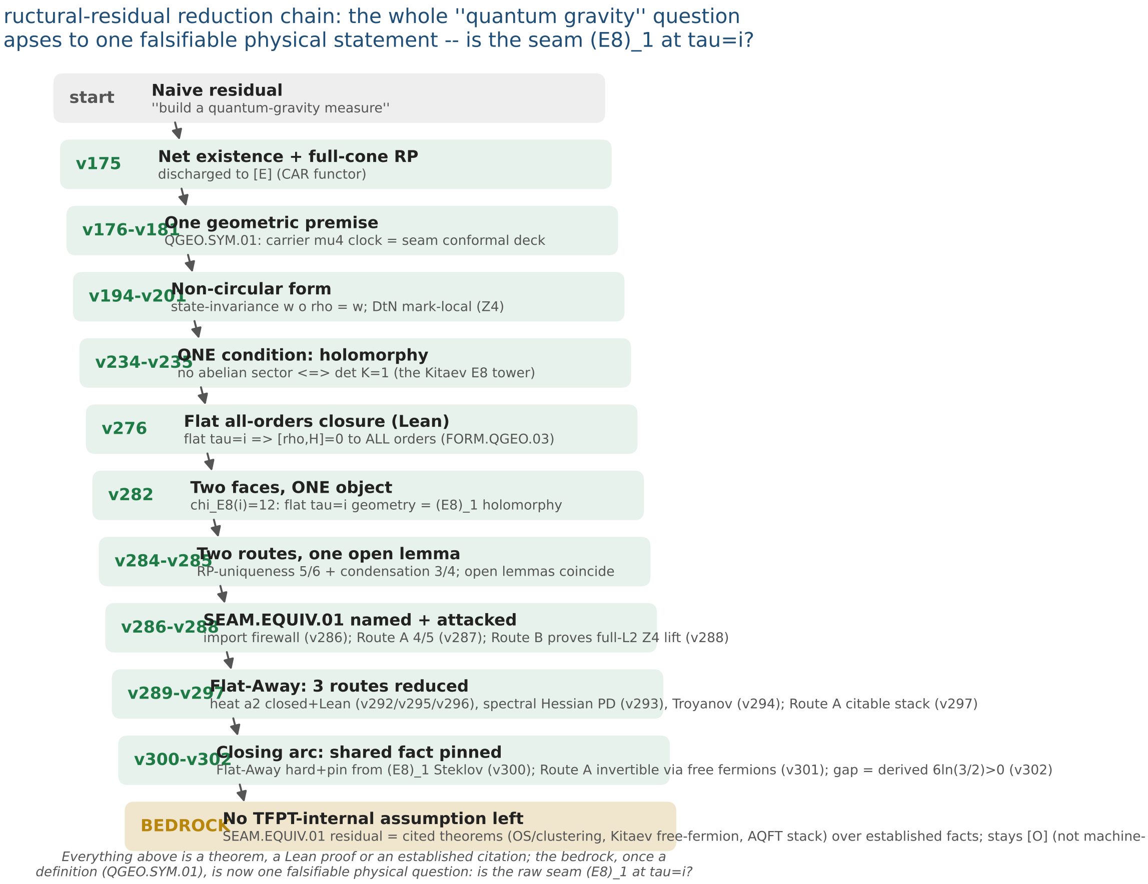

Figure 3: The structural-residual reduction chain (v175–v181). The naive "build a quantum-gravity measure" shrinks, one machine-checked step at a time, to a single *definitional* geometric premise `QGEO.SYM.01` (the carrier μ_4 clock is the seam's conformal deck). Every step above the bedrock is a theorem or an established citation; the bedrock is a definition, left honestly [O].

integer condition is the physics of topological order: a K-matrix phase has ground-state degeneracy $|\det K|^g$ on a genus-g surface, so $\det K = 1 \Leftrightarrow$ *no* topological ground-state degeneracy on any closed surface $\Leftrightarrow$ the bulk is *short-range-entangled* (invertible). Among rank-8 even positive-definite K ($c = 8 = g_{\text{car}} + N_{\text{fam}}$) only E_8 is SRE — the unique nontrivial bosonic SRE 2+1d phase, the Kitaev E_8 state, whose edge is the holomorphic $(E_8)_1$. TFPT already has a unique vacuum on the plane (RP + gap + cluster + OS); the plane cannot see $\det K$ (no non-contractible cycles), the torus can. So the one open analytic step is the sharp, falsifiable physical statement *"the free RP gapped seam bulk is short-range-entangled (no topological order)"* (= $\det K = 1$ = the Kitaev E_8 phase, `GATE.HOLO.03`, [E]/[O]) — a yes/no question about topological order on the seam, sharper than the definitional `QGEO.SYM.01`.

Lemma 2 (U1 — wall landscape, *corrected*)**.** *Parabolic degree zero forces the line weight-sums onto the wall* $(w_1, w_2, w_3) = (2, 1, 1)$*. An explicit balanced representative (weights in* $\frac{1}{3}$*) is*

$$W_{\text{wall}} = \frac{1}{3}\begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of $(0, 1, 2)$*, row sums* $(2, 1, 1)$*.* ***Honest result:*** *the* 6^4=1296 *column-permutation matrices give* 144 *walls; under the* full *group* $D_4 \times S_3(\textit{lines}) \times S_3(\textit{values})$ *they form* five *orbits,* not one*. So* W_{wall} *is* not *singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite,* *explicitly enumerated set; $W_{\rm wall}$ is one of the five. Machine: fully certified $C_U^{(1)}$ (*[`verification/v29_research_contract_certs.py`]*).*

The splitting type is now algebraic (RH route, [`verification/v32_rh_splitting.py`]***).*** *Passing from the monodromy to the Fuchsian* residues *A_k (eigenvalues = weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, $A_k = U^{k-1}A_0U^{-(k-1)}$, $U = \mathrm{diag}(1, i, -i)$) the twisted $\mathbb{Z}_4$-average collapses* exactly*: $\sum_k U^kA_0U^{-k} = 4\,\mathrm{diag}(A_0)$, so the exponents at infinity are $4\,\mathrm{diag}(A_0)$. A flat bundle with regular ∞ needs integer exponents summing to $4 = -\deg E$; with the cusp weights the only options are perms of $(2, 1, 1)$ and $(2, 2, 0)$, i.e.*

$$\boxed{\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2 \iff \mathrm{diag}(A_0) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4})}$$

(the sibling $(2, 2, 0) \to \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$). Such a Hermitian A_0 (spectrum $\{0, \frac{1}{3}, \frac{2}{3}\}$, that diagonal) exists by Schur–Horn. So the splitting *is an algebraic condition on* $\mathrm{diag}(A_0)$. Honest wall: *the global $\prod M_k = \mathbf{1}$ is the path-ordered monodromy (*not $\exp(2\pi i \sum A_k)$*), constraining* $\text{off-diag}(A_0)$ *via the genuine RH solve; and R-extraction still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ bridge. The RH route fixes the* splitting *algebraically but not c_u/c_d.*

Lemma 3 (U2 — the kill switch, *run and hardened* [E])**.** *The selected polystable wall representative must* not *collapse to the diagonal abelian $U(1)^3$ point if it is to give non-trivial $SU(3)_F$ transport mixing:* ***(A)*** *$\nabla_F^\star$ polystable but* not *simultaneously diagonal (the desired breakthrough), or* ***(B)*** *diagonal collapse, in which case c_u/c_d stays honestly [C].* ***Result: the cheap test is run robustly and lands on A.*** *The kill-switch fires iff the D_4-symmetric cusp representation is* reducible *(commutant non-scalar). Across* 400 *random cusp-class generators of the $\mathbb{Z}_4$ family $M_k = U^{k-1}MU^{-(k-1)}$ the commutant is scalar at* every *point (*irreducible 400/400*), the reducible locus is hit with probability* 0 *(measure-zero, codimension ≥ 1), and the mixing strength stays bounded away from zero (*min *off-diag* $= 0.35 > 0.05$*). So case A is the* generic *behaviour, not an accident of the one explicit point of* [`verification/v33_explicit_flat_bundle.py`]*:* $(U_{\rm wall})$ survives *its own cheapest falsification test.* [`verification/v38_uwall_killswitch.py`]

Lemma 4 (U3 — D_4 fixed locus as an explicit variety, *positive-dimensional*)**.** *With $M_k = U^{k-1}AU^{-(k-1)}$ ($U = \mathrm{diag}(1, i, -i)$, the $\mathbb{Z}_4$ rotation) and the reflection $M_1 = VM_1^{-1}V^{-1}$, $M_3 = VM_3^{-1}V^{-1}$, $M_2 = VM_4^{-1}V^{-1}$ (with $V = -[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$, $V^2{=}1$, $VUV^{-1}{=}U^{-1}$), the relations become polynomials in the entries of $A \in C_{\rm cusp}$; $\mathcal{X}^{D_4}_{\rm cusp}$ is cut out as $\bigcup_k C_k$, parametrised by trace coordinates $x = \mathrm{tr}(M_1M_2)$ (note $\mathrm{tr}\,A = 1{+}\omega{+}\omega^2 = 0$).* ***Result ($C_U^{(2)}$,*** [`verification/v30_d4_character_variety.py`]***):*** *adding the reflection to the $\mathbb{Z}_4$ locus of v19 does* not *isolate the point — the* full *D_4-fixed product locus is still* positive-dimensional *($|\mathrm{tr}(M_1M_2)|$ varies continuously over $\approx [0, 3]$). So even the full D_4 symmetry cannot select $\nabla_F^\star$.* Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

Lemma 5 (U4 — selectors as equations)**.** *The readouts $\rho \mapsto \det R(\rho)$, $\rho \mapsto \mathrm{Spec}(Q_+(\rho))$, $\rho \mapsto \Lambda_{f,j}(\rho)$ are well-defined (cyclotomic/algebraic) on $\mathcal{X}^{D_4}_{\rm cusp}$.* ***Acceptance criterion:***

$$\#\{\rho \in \mathcal{X}^{D_4}_{\rm cusp} :\ \det R = 8,\ \mathrm{Spec}(Q_+) = \{1, 2, 3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not *closed.* ***The bottleneck (made precise by U3):*** *since the D_4-fixed variety is positive-dimensional, the* selector *must do all the cutting — and evaluating* $\det R(\rho)$, $\mathrm{Spec}(Q_+(\rho))$ *as functions on $\mathcal{X}^{D_4}_{\rm cusp}$ requires the* parabolic-degree ↔ residue dictionary *$R(\rho)$, i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the $(U_{\rm wall})$ gate.* Status update: since v118–v122 the dictionary itself is exact *— hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the $W(A_3)$ monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller R4′, the* establishment of the selectors *($n = (5, -9, 6)$ + the diamond determinants; since* [`verification/v134_dual_anchor.py`]*: the pair (d, n)).*

What $R(\rho)$ ***is*** ***(*** *[verification/v31_R_dictionary.py]* ***).*** *Two facts pin its nature.* (i) Case A is alive: *every* D_4*-fixed cusp monodromy found is* irreducible*, so the wall point carries non-trivial* $SU(3)_F$ *mixing — the U2 kill switch does* not *fire to the diagonal case B.* (ii) $R(\rho)$ is not algebraic: R *is integer-valued, but the algebraic invariants of* ρ *(e.g.* $\mathrm{tr}(M_1M_2)$*) vary* continuously *on the locus; an integer readout cannot be a continuous algebraic function. Hence* $R(\rho)$ *is the discrete invariant of the* non-abelian Hodge / Mehta–Seshadri *map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle* E_ρ*): integer output, but its* computation *is the transcendental harmonic-metric (Hitchin) solve* on the generic Higgs locus. ***Refinement (U4***$''$***,*** *[verification/v40_harmonic_metric.py]* ***):*** *at the* polystable *point TFPT actually selects this pessimism does not apply — see the box below.*

Progress: the *selector side* of U4 is closed on the explicit point ([E]). While $R(\rho)$ as a function on the whole locus needs the Hitchin solve, the two U4 *selectors* are *algebraically* accessible on the explicit flat bundle of [verification/v33_explicit_flat_bundle.py] — no PDE. Computed exactly from A_0: the exponents at infinity $\mathrm{eig}(\sum_k A_k) = \{2,1,1\}$ give the splitting type $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, i.e. the parabolic *anchor* $a = (1,1,2)$; the parabolic degree is 0 ($\deg E = -4 = -|\mu_4|$ plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\boxed{\begin{array}{c}\det R = 8 = n\cdot a \quad \text{(anchor pairing on the splitting type)},\\ \mathrm{Spec}(Q_+) = \{1,2,3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\})\end{array}}$$

So $\det R = 8$ is the anchor pairing of the splitting type and $\mathrm{Spec}(Q_+)$ is the affine image (d=3 the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle's algebraic data*. **Residual (sharpened below by Readout Rigidity):** the quark *ratio* c_u/c_d turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation (U_{point}), which is an anchor, not a missing PDE. [verification/v39_uwall_selectors.py]

Breakthrough: the harmonic metric is *finite linear algebra*, not a PDE ([E]). The U4 "transcendental Hitchin solve" is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field $\Phi = 0$. So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [verification/v33_explicit_flat_bundle.py] this is realised: the raw monodromies M_k are non-unitary ($\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$), but there is a *unique* common invariant Hermitian form H ($\dim\ker\{M_k^\dagger H M_k = H\} = 1$), it is *positive-definite*, and the unitarised holonomy $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$ is unitary ($\sim 10^{-8}$), lies in the cusp class $\{1, \omega, \omega^2\}$ with $\prod \widetilde{M}_k = \mathbf{1}$, and has *clean* matrix elements $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$. **So** $C_U^{(3)}$ **reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length** R **(closed: H1 +** $\det R = 8$**).** *Residual:* the feared PDE is gone; the quark *ratio* c_u/c_d is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [verification/v40_harmonic_metric.py]

The final leg-assignment test (honest result) ([E]). Does the harmonic-frame holonomy $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ feed the $\delta = \frac{1}{2}$ resolvent to give the lepton amplitudes $\Lambda = (0.475, 1.107, 0.917)$? **(i) Ruled out as a literal identity:** $\Lambda_\mu = 1.107 > 1$, while unitary holonomy entries have modulus ≤ 1, so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly $|\mathrm{diag}\,\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, and that $\frac{1}{2}$ *equals* the distinguished lepton transport value $\delta = \frac{1}{2}$ that v20 used by hand — so the geometry plausibly *supplies* δ (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g. $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the $\delta = \frac{1}{2}$ resolvent (closed combinatorially via the word-length R);

the holonomy supplies δ and the leg selection, not the amplitude magnitude. c_u/c_d is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ($\delta = \frac{1}{2}$). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [`verification/v41_leg_assignment.py`]

The torsion normal form and the $\delta = \frac{1}{2}$ theorem ([E]). The v41 "genericity is future work" item is now executed — and yields a theorem. *(i) Flatness = μ_4 torsion [E]:* for *any* M (symbolic), $\prod_k U^k M U^{-k} = (MU)^4$ — the $\mathbb{Z}_4$-family flatness constraint *is* the torsion statement "$T = MU$ is a fourth root of unity": the μ_4 atom appears as the *order* of the twisted cusp generator. *(ii) The $\delta = \frac{1}{2}$ theorem [E] (both directions):* on the *involutive* branch ($T^2 = \mathbf{1}$) T is a Hermitian reflection $2vv^\dagger - \mathbf{1}$, and the cusp trace condition $2v^\dagger U^{-1} v = 1$ splits *exactly* into $|v_1|^2 = \frac{1}{2}$ (real part) and $|v_2| = |v_3|$ (imaginary part), forcing

$$\boxed{\operatorname{diag} M = (0, \tfrac{i}{2}, -\tfrac{i}{2}), \qquad \operatorname{spec} M = \{1, \omega, \omega^2\} \text{ automatic}}$$

($\chi_M = \lambda^3 - 1$ from unitary + trace 0 + det 1). So $\delta = |M_{22}| = \frac{1}{2}$ holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a *μ_4-torsion identity*, not an observation. *(iii) Reflection lemma [E]:* the D_4 reflection forces $\operatorname{diag} M = (r, z, \bar{z})$ with r real and $\operatorname{Re} z = -r/2$ — the whole D_4 diagonal freedom is *one* real parameter, and $\delta = \frac{1}{2} \iff r = 0$. *(iv) Branch census [E] (seeded):* the full D_4-fixed flat cusp locus realises only $\operatorname{tr}(MU) \in \{1, -1\}$; the involutive branch has the diagonal *constant* $(0, \frac{1}{2}, \frac{1}{2})$ to machine precision, while on the $\{1, i, -i\}$ branch d_1 varies over $[0, 1]$ — and the explicit v33/v40 point sits exactly in its $d_1 = 0$ slice, the *same* δ value. *Residue [C] (recorded):* whether the anchor splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (Mehta–Seshadri side) forces the $d_1 = 0$ slice — or selects the involutive branch — is the remaining question; $\delta = \frac{1}{2}$ itself is no longer the open item. [`verification/v114_torsion_delta.py`]

The anchor pins the residue: an exact normal form for the (U_{wall}) bundle ([E]). The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form*. *(i) μ_4-average lemma [E]:* $\sum_k U^k X U^{-k} = 4 \operatorname{diag} X$ for any X — μ_4 conjugation-averaging *is* the diagonal projection, so the exponents at infinity are $4 \operatorname{diag} A_0$ and the anchor splitting holds *iff* $\operatorname{diag} A_0 = (2, 1, 1)/4 = a/|\mu_4|$: the residue diagonal **is** the anchor over μ_4 (v33's diagonal was forced, not chosen). *(ii) The $(8, 0, 5)/144$ lemma [E]:* the anchor diagonal and the cusp spectrum $\{0, 1, 2\}/N_{\text{fam}}$ pin $\sum |a_{ij}|^2 = \frac{13}{144}$, and with $a_{13} = 0$ the determinant condition solves *uniquely* to

$$\boxed{(|a_{12}|^2,\ |a_{13}|^2,\ |a_{23}|^2) = \frac{(8,\ 0,\ 5)}{144}} \qquad \begin{array}{l} 8{=}\operatorname{rank} E_8,\ 5{=}g_{\text{car}}, \\ 144{=}(|\mu_4| N_{\text{fam}})^2,\ 8{+}5{=}13{=}\Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same* Δ_Q that denominates $c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$. *(iii) Exact normal form [E]:* $A_0^\star = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$ has $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ *exactly*. *(iv) $A_0^\star$ is the RH solution [E]:* its big-circle monodromy is trivial to 10^{-10}, $\prod M_k = \mathbf{1}$, and the unitarised holonomy has $|\operatorname{diag} \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, $\operatorname{tr}(\widetilde{M}_0 U) = 1$ — the v33/v40 point is gauge-equivalent to $A_0^\star$ (the a_{12}, a_{23} phases are diagonal gauge). *(v) The anchor forces the $d_1 = 0$ slice [E]:* a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with $|\operatorname{diag} \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**; $\delta = \frac{1}{2}$ needs no selection. *Residue [C]:* $a_{13} = 0$ and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining U_{point} step. [`verification/v115_anchor_residue.py`]

The resonance theorem: $M_\infty = \mathbf{1} \iff a_{13} = 0$ — one gauge orbit [E] ([`verification/v116_resonance_uniqueness.py`])

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as $dY/dw = -(1/w)(B_0 + B_1 w + \cdots)Y$ with $B_m = \sum_k i^{km} U^k A_0 U^{-k}$, the *twisted* μ_4

averages collapse **[E]**:

$$B_0 = 4\,\mathrm{diag}\,A_0, \qquad \boxed{B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}}$$

(only the eigenvalue-ratio $-i$ cells survive). The anchor exponents $\mathrm{diag}(2,1,1)$ have resonance gap 1; the level-1 formal-gauge equation $(1-\mathrm{ad}\,R_0)H_1 = R_1$ is singular exactly on the cells $\{(2,1),(3,1)\}$, where B_1 has entries $(0,\, 4\overline{a_{13}})$; all higher levels are non-resonant and the formal monodromy is $e^{-2\pi i\,\mathrm{diag}(2,1,1)} = \mathbf{1}$. Hence **[E]**:

$$\boxed{M_\infty = \mathbf{1} \iff a_{13} = 0}$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation*. *Uniqueness corollary [E]:* $a_{13} = 0$ + the $(8,0,5)/144$ moduli + phases = diagonal gauge ⇒ the flat anchor locus is **exactly one gauge orbit**, the exact $A_0^\star$: within the μ_4-equivariant anchor class the (U_{wall}) bundle datum is *algebraically unique*. *Sharpness [E]:* $\|M_\infty - \mathbf{1}\| \sim 8.5\,|a_{13}|$ (0.17 at $a_{13} = 0.02$) vs 10^{-10} at $A_0^\star$. *Honest scope [C]:* the equivariant ansatz is the established D_4 selector input (v19/v30); the *residue* side is now **[E]**, while the holonomy values (the harmonic diagonal $(0, \frac{1}{2}, \frac{1}{2})$) remain **[E]** — transcendental monodromy of an exact system.

The monodromy is the Weyl group of A_3 — the holonomy is exact [E] ([`verification/v117_monodromy_weyl_a3.py`])

The last **[E]** item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact $A_0^\star$ system is itself *exact*, with entries in $\frac{1}{2}\mathbb{Z}[i]$:

$$\boxed{\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}}$$

unitary, $\det = 1$, $\mathrm{tr} = 0$, $\chi = \lambda^3 - 1$ (the cusp class *exactly*), $\widetilde{M}_0^3 = \mathbf{1}$, and $\mathrm{diag}\,\widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$ — so $d_1 = 0$ and $\delta = \frac{1}{2}$ are now *matrix entries*, not observations; $(\widetilde{M}_0 U)^4 = \mathbf{1}$ with $\mathrm{tr}(\widetilde{M}_0 U) = 1$ (the $\{1, i, -i\}$ branch, $d_1 = 0$ slice — exactly where v114/v115 located the physical point). *The group* **[E]**: exact enumeration of $\langle U, \widetilde{M}_0\rangle$ gives order 24 with order statistics $(1,9,8,6)$ and character values $(3,-1,0,1)$ — uniquely S_4 in its standard ⊗ sign representation, i.e. the (twisted) reflection representation of the **Weyl group of the family lattice** A_3:

$$\boxed{\mathrm{monodromy}(U_{\mathrm{wall}}) = W(A_3) = S_4, \qquad |W(A_3)| = 4! = 24}$$

— the flavor wall realises the A_3 factor of the glue $D_5 \oplus A_3 + \mu_4$ as its Weyl group (U = a 4-cycle, the μ_4 deck; $\widetilde{M}_0$ = a 3-cycle, the family rotation). *Identification [E]:* the ODE monodromy of $A_0^\star$ equals the exact $\widetilde{M}_0$ to 3.5×10^{-10}, and the invariant form H is U-invariant to 10^{-11}. *Status:* holonomy values **[E]**; only the analytic identification stays **[E]**. The δ thread (v20 → v40/v41 → v114 → v115 → v117) is closed end to end.

The hexagon is the sign-twisted family spectrum — first exact H2 piece [E] ([`verification/v118_hexagon_family_dictionary.py`])

The missing "$C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary" (H2) acquires its first exact entry. *Sign-twist lemma [E]:* $-\widetilde{M}_0$ has order 6 and

$$\boxed{\mathrm{spec}(\widetilde{M}_0) \cup \mathrm{spec}(-\widetilde{M}_0) = \mu_6}$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ($\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ realised as $(-1) \times \langle\widetilde{M}_0\rangle$); the v20 denominators $\frac{5}{4} - \cos(r\pi/3)$ are exactly the eigen-denominators $|1 - \zeta_6^r/2|^2$ of the two family resolvents $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$. *Cyclotomic determinant [E]:* $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1-t)\Phi_3(t)$, so $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$ — and the lepton coefficients become resolvent determinants: $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$, $c_\mu = N_{\mathrm{fam}}/(2\det_+) = \frac{4}{3}$, $c_\tau = |\mu_4|\det_-/N_{\mathrm{fam}} = \frac{7}{6}$, product rule $= |\mu_4|/\det_+ = \frac{32}{9}$. The lepton 7 and 9 *are* the family-resolvent determinants; $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$ is supplied by the matrix itself. *Sheet-extended group [E] (audit):* $\langle U, \widetilde{M}_0, -\mathbf{1}\rangle$ has order $48 = \Omega_{\mathrm{adm}} = N_{\mathrm{fam}}\dim S^+$ — the seed's quartic coefficient. *Residue [C]:* the dictionary's *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20's established input / open — nothing re-fished.

The address table: the lepton words are the compiler atoms [E]/[C] ([verification/v120_address_table.py])

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today's readings (the red-team firewall). *(i) The lepton words are the atoms [E]:*

$$\boxed{(L_e, L_\mu, L_\tau) = (8,5,3) = (\operatorname{rank} E_8,\ g_{\text{car}},\ N_{\text{fam}}) = (p_0{+}e_2,\ e_2,\ p_0)}$$

in mass order (longest word = lightest lepton). *(ii) Addresses = division by the hexagon [E]:* $(r,w) = L \operatorname{divmod} p_2$ with $p_2 = 6 = |R^+(A_3)|$ itself an atom: $e \to (2,1)$, $\mu \to (5,0)$, $\tau \to (3,0)$; $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$, $\sum L = 16 = \dim S^+$. *(iii) Sheet parity [E]:* by the v118 dictionary, e sits on the *untwisted* sheet ($\omega \in \operatorname{spec} \widetilde{M}_0$), μ, τ on the *twisted* sheet $(-\omega, -1)$; the anchor lepton τ occupies the unique *real* twisted eigenvalue -1 — exactly the site where v20's product rule replaces the resolvent. *(iv) Quark sum rules [E] (frozen v18 words* $7,7,5,3,2,0$*):* up-type sum $= 10 = p_3$; down-type sum $= 14 = p_1{+}p_3$; **all quarks** $= 24 = |W(A_3)|$ (the monodromy group order, v117); **all nine charged fermions** $= 40 = p_1p_3 = a^TRa$ (v119); quark $\sum r = 2p_2$, $\sum w = |\mathbb{Z}_2|$; the top sits at the vacuum site $(0,0)$ — zero word, no suppression, the established ladder anchor. *Residue [C]:* five independent closures $(16, 10, 14, 24, 40)$ constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

The address table is pinned: R is the unique hexagon matrix with atom margins [E] ([verification/v121_address_pinning.py])

The per-fermion assignment closes at the *information* level. The established identity $L = R{+}2U$ (v95) says the word-length table *is* L[sector][generation] — rows (up; down; lepton) $= ((7,3,0);(7,5,2);(8,5,3))$, the v18/v20 words verbatim — so the addresses are the entries of the *residue operator* R plus one hexagon turn for the first generation, and R's first column is the **anchor** $a = (1,1,2)$. The margins are atoms [E]: row sums $(4,8,10) = (e_1, \operatorname{rank} E_8, p_3)$, column sums $(4,13,5) = (e_1, \Delta_Q, g_{\text{car}})$. And the census closes it [E]:

$$\boxed{\begin{array}{c}\text{among all } 3\times 3 \text{ matrices with entries in the hexagon } \{0..5\},\\ \text{exactly 17 have the atom margins — and exactly ONE has } \det = 8\end{array}}$$

— and it is R (also the unique one with SNF $= (1,1,8)$; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ($m \sim \lambda_Y^L$: longer word = lighter fermion). *Residue [C]:* the open H2 item shrinks from "derive 9 per-fermion integers" to "derive the six atom margins + det = rank". *Firewall:* R, L and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

The margins are theorems ([verification/v122_margin_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — *(S1)* the D_4 annihilator $n = (5,-9,6)$ with $n^TR = (8,0,0)$ (it kills the unchanged (c_2, c_3) plane; $n\cdot a = 8$), *(S2)* $\det R = 8$, *(S3)* $\det K = 4$ with $K = R + Q\Sigma$ — pin R *uniquely* among hexagon matrices [E]: the $(S1)$ census has $4\times4\times4 = 64$ candidates, $(S2)$ leaves 12, $(S3)$ leaves **exactly one**, and it is R. Hence the atom margins, the anchor column $Re_1 = a$ and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]:* on the $(S1)+(S2)$ locus $\det(M+2U) = 20 = \det L$ holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [C]:* H2 reduces to the establishment of the selectors themselves (n and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class.*

The dual-normal selector: (d,n) pins R columnwise ([verification/v136_dual_normal_selector.py]). The selector list shrinks once more. With the dual anchor $d = a^\top R^{-1}$ ([verification/v134_dual_anchor.py]) and the torsion normal n, each *column* of R obeys $d\cdot c_j = a_j$ and $n\cdot c_j = 8\delta_{1j}$; the common kernel of (d,n) is the primitive vector $(6,8,7)$ (audit: p_2, $\operatorname{rank} E_8$, scalaron-7), so within the address box $\{0..5\}^3$ each column is the *unique* lattice point (census: 1 of 216 each) — $(d,n) \Rightarrow R$, and the 6^9 census above collapses to three column problems [E]. *Honesty:* K is *not* pinned directly ($d^\top K = (1, \frac{1}{2}, 1)$); $\det K = 4$ comes via the μ_4 wall ([`verification/v135_det_surface.py`]). *The mechanism is bounded:* the zero mode of $A_0^\star$ carries the exact spectral normal $\sigma = (2, -9, 5) = (|\mathbb{Z}_2|, -N_{\mathrm{fam}}^2, g_{\mathrm{car}})$ (signed squares; $\|v\|^2 = \frac{16}{9}$ with $16 = \dim S^+$, $\operatorname{adj} A_0^\star = \frac{2}{9} P_v$ with $\frac{2}{9}$ = the SdS entropy curvature), $\sigma \cdot \mathbf{1} = -|\mathbb{Z}_2|$, $\sigma \cdot a = N_{\mathrm{fam}}$, $n \cdot \sigma = 121 = \|\operatorname{Pl}(K)\|_1^2$; and the exact $W(A_3)$ orbit control shows *no* group image of σ is proportional to n — the naive cofactor/orbit lift fails, the step $\sigma \to n$ is a genuine mechanism beyond the monodromy group. *R4′ restated [C]:* derive the *pair* (d,n); then R is forced columnwise.

The selector triangle ([`verification/v139_selector_triangle.py`]). The pair itself now collapses further. First, d is *pure anchor data:* $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ exactly — the closed form never mentions R, so the first selector is *derived*, not residual. Second, the frame $(\mathbf{1}, a, \sigma)$ is a basis of $\mathbb{Z}^3$ with $\det(\mathbf{1}|a|\sigma) = 11 = \|\operatorname{Pl}(K)\|_1$ (the quark constant as the frame volume), and n is the *unique* integer covector with the three atom pairings

$$\boxed{n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2|, \qquad n \cdot a = 8 = \operatorname{rank} E_8, \qquad n \cdot \sigma = 121 = \|\operatorname{Pl}(K)\|_1^2}$$

— the cofactor definition is replaced by frame data plus three atoms. On the $(\mathbf{1}, a)$-constrained line the σ-pairing is $11(11{+}t)$: divisible by 11 for *every* t, a perfect square exactly at the true n ($t{=}0$). The review's predicted lift is exact as an audit identity: $n = \operatorname{reverse}(\sigma) + |\mu_4| e_3$ (generation flip + μ_4 lift at the third slot).

Frame integrality: three pairings → two ([`verification/v142_frame_integrality.py`]). The third pairing is *not* independent. The pairing triples $(m \cdot \mathbf{1}, m \cdot a, m \cdot \sigma)$ of *integer* covectors form an index-11 sublattice of $\mathbb{Z}^3$, cut out by the single congruence

$$\boxed{x - 3y + z \equiv 0 \pmod{11}} \qquad (\text{index } 11 = \|\operatorname{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms $(x,y) = (|\mathbb{Z}_2|, \operatorname{rank} E_8) = (2,8)$, integrality *forces* $z \equiv 0 \pmod{11}$ — the $11(11{+}t)$ line *is* the integer line. Primitivity forces t *even* (odd t makes every entry of $n(t)$ even), and $t{=}0$ is the unique even t with a square σ-pairing for $|t| < 88$ (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities ($n \cdot a = \det R$ by Laplace along the anchor column; $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$ with $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$, v134) — restructuring, not establishment.

The pairing values are atom identities ([`verification/v145_pairing_atoms.py`]). Even the two line-pairing *values* carry no independent information. Reading the v139 lift structurally — $n = w_0(\sigma) + |\mu_4| e_3$ with w_0 the A_3 *exponent duality* $m \leftrightarrow h{-}m$ (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n{\cdot}\mathbf{1} = |\mu_4| {-} |\mathbb{Z}_2| = |\mathbb{Z}_2|, \qquad n{\cdot}a = |\mu_4|\, a_3 = 8 = \operatorname{rank} E_8 \ \text{ via } \ \boxed{|\mu_4| + g_{\mathrm{car}} = N_{\mathrm{fam}}^2 \quad (4 + 5 = 9,\ \sigma \perp w_0(a))},$$

and $n \cdot \sigma = \|\sigma\|^2 - N_{\mathrm{fam}}^2 + |\mu_4| g_{\mathrm{car}} = 110 - 9 + 20 = 121$ with $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\mathrm{fam}}^4 + g_{\mathrm{car}}^2 = 110 = |\mathbb{Z}_2| g_{\mathrm{car}} \|\operatorname{Pl}(K)\|_1$. *R4′ residue [C] (final form):* derive the *one* map $n = w_0(\sigma) + |\mu_4| e_3$ — no independent number remains; everything else in the chain anchor $\to (d,n) \to R$ is exact.

Both dual normals are anchor data ([`verification/v149_cusp_normal.py`]). The remaining opacity dissolves in the canonical frame: the pairings of n with the *cusp eigenbasis* (the geometric basis of v98) are exactly

$$\boxed{n \cdot e_1 = 6 = p_2(a), \qquad n \cdot e_2 = 3 = p_0(a), \qquad n \cdot e_3 = 5 = e_2(a)}$$

— one anchor invariant per Q_+ degree, and the cusp matrix is unimodular, so these three pairings determine n uniquely. With $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal *pair* (v134) is anchor data through and through — one normal per frame; the cusp data $(6,3,5)$, the frame pairings $(2, 8, 121)$ and the w_0-lift are three presentations of one covector, all atomic (equivalences exact). *Audit:* $\sigma \to (5,1,2)$; alternate reading $(6,3,5) = (|R^+(A_3)|, N_{\mathrm{fam}}, g_{\mathrm{car}})$; sum $14 = \dim G_2$. *R4′ residue [C] (cusp form):* derive the *assignment* — why the Q_+ degrees $(1,2,3)$ carry (p_2, p_0, e_2) in that order; one discrete assignment, zero continuous freedom.

Exterior Leg Lemma — the quark u/d leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The v41 negative is a *category error*, not a dead end. u and d share the hexagon-radial data $(r, w) = (1, 1)$ ($L = 7$ for both), and the C_6 resolvent $c(r, w) = |\mu_4|^w/(\frac{5}{4} - \cos\frac{r\pi}{3})$ reads *only* (r, w) — so any *scalar* leg gives $c_u/c_d \equiv 1$. A scalar sees radius, not orientation. The u/d separation lives in the smallest *non-scalar* invariant of the $SU(3)_F$ composition: the exterior square $\Lambda^2 F$, i.e. the oriented anchor-plane area $\mathrm{Pl}_{\mathbf{1},a}(K)$. The "11" is then *generated*, not read:

$$(K, \mathbf{1}, a) \to B_K = \left(\begin{smallmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{smallmatrix}\right) \to \mathrm{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \to \|\,\mathrm{Pl}(K)\|_1 = 11,$$

$$\boxed{\frac{c_u}{c_d} = \frac{g_{\mathrm{car}}\,\|\,\mathrm{Pl}_{\mathbf{1},a}(K)\|_1}{N_{\mathrm{fam}}^2\,\Delta_Q} = \frac{5\cdot 11}{9\cdot 13} = \frac{55}{117}} \quad \left(\text{anchor microcode } = \frac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}\right).$$

Typing: [E] for the Plücker / microcode identities ($\mathrm{Pl}(K) = (-1, 6, 4)$, $\|\cdot\|_1 = 11$, $5{\cdot}11/(9{\cdot}13) = 55/117$); and — by Readout Rigidity (v49, below) — [E] also for the *physical* ratio $\mathcal{C}_{ud}(\rho^\star)$ = this $\Lambda^2 F$ readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on $\Lambda^2 F$. The quark residual is thereby re-typed from "find the right scalar y" to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale (U_{point}) stays as an anchor, no new scalar, no fishing.

Audit (the 121 *lemma,* [`verification/v119_review_validation_2.py`]*):* the "11" has a second, independent appearance [E]: with $h = \mathbf{1} + a = (2, 2, 3)$ in the canonical anchor ordering, $h^T R h = 121 = 11^2 = \|\,\mathrm{Pl}(K)\|_1^2 = (p_3 + 1)^2$ — the quadratic *anchor norm* of the residue operator R reproduces the Plücker integer (sensitivity control: the three orderings of h give $\{105, 121, 135\}$; only the canonical one yields 121). Bonus checksums: $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$ (the entry sum of R) and $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$. Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [`verification/v42_exterior_leg.py`]

Bridge test ([`verification/v43_exterior_bridge.py`]). The typing is confirmed: for the $SU(3)$ holonomy the exterior square is the conjugate representation, $\Lambda^2(\widetilde{M}) = \widetilde{\widetilde{M}}$ exactly — the exterior leg *is* the $\bar{\mathbf{3}}$. But the *continuous* action $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$ does *not* equal the *integer* $\mathrm{Pl}(K) = (-1, 6, 4)$: the integer Plücker is the combinatorial K datum. So the bridge $\rho^\star \to \mathrm{Pl}(K)$ is the *discrete* non-abelian-Hodge invariant (the integer R; $\det R = 8$ and $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$ already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the $\Lambda^2 F$ side.

Lie-level grounding — no special math ([`verification/v44_carrier_exterior.py`]). The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier, $16 = \Lambda^{\mathrm{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$, and under $SU(5) \subset SO(10)$ the up quark sits in $10 = \Lambda^2(5)$ while the down sits in $\bar{5} = \Lambda^4(5)$. So the *exterior degrees* are $\deg(u^c) = 2$, $\deg(d^c) = 4$, with $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$ (the hexagon) and $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$ (the sheet). The "Λ^2" of the exterior leg is therefore not exotic Hodge data — it is the carrier's own Λ^2 grading (the up quark *lives* in $\Lambda^2(5)$). This re-types the residual once more: from "discrete non-abelian-Hodge invariant" to "the carrier exterior degree" — a standard [E] fact. *Honest scope:* this grounds the *type*; the Plücker *number* 55/117 stays the family $\mathrm{Pl}(K)$ [E], and the carrier-Λ^2 ↔ family-$\Lambda^2 F$ link (via family ⊗ carrier = 15 = 16−1) plus the normalisation remain [C].

The "11" is Pascal — a third, simplest origin ([`verification/v45_family_exterior.py`]). Following that link, $\mathrm{Pl}(K)$ is the *exterior algebra of the family fundamental* $4 = \mu_4$ (the $A_3{=}SU(4)$ fundamental): $\Lambda^k(4) = \binom{4}{k}$ is Pascal row $4 = (1, 4, 6, 4, 1)$ with full sum $2^4 = 16 = \dim S^+$, and $|\,\mathrm{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$ are exactly its distinct exterior powers. Hence

$$\boxed{\|\,\mathrm{Pl}(K)\|_1 = \textstyle\sum_{k=0}^{2} \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\mathrm{car}}}$$

(the cumulative $\Lambda^\bullet(\mu_4)$ up to degree $2 = \deg(u^c)$), and family ⊗ carrier $= 15 = \dim\mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10{+}5 = N_{\mathrm{fam}} g_{\mathrm{car}}$ read three ways. So the "11" is generated a *third* way — the cumulative family exterior algebra up to the up-quark's degree — pure Pascal/branching, *no special math*. The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

Lemma 6 (U5 — the "11" worry, *discharged* by Readout Rigidity)**.** *The original worry was that* $c_u/c_d = \mathcal{C}_{ud}(\rho^\star)$ *must be evaluated at the uniquely selected* $\rho^\star$ without *using* 11 *as input, so that until then* 55/117 *would be a mere table-level rational shadow.* ***This is now resolved (v49):*** $\mathcal{C}_{ud}$ *is* constant *on the discrete selector stratum* $\mathcal{S}_{8,Q}$*, so the "11" is* earned *as the rigid Plücker* *readout* $\|\,\mathrm{Pl}(K)\|_1$ *(v42) —* independently *of point-uniqueness. Point-uniqueness of* $\rho^\star$ *enters only the* absolute *amplitude scale, not the ratio.*

Lemma 7 (U6 — the four certificates)**.** $C_U^{(1)}$ *wall enumeration (done);* $C_U^{(2)}$ D_4*-fixed character solver;* $C_U^{(3)}$ *selector uniqueness (one* S*-equivalence class);* $C_U^{(4)}$ *amplitude extraction — the* ratios $R, Q, c_u/c_d$ *are read off the (derived) selector stratum (closed, v49/v71); only the* absolute $U_f^\star, c_q$ *normalisation is the anchor.*

Progress: an explicit valid flat bundle is realised (RH solve) ([**E**]). A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue A_0 (eigenvalues $\{0, \frac{1}{3}, \frac{2}{3}\}$, diag $= (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$) for which the Fuchsian system $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)}/(z - i^k)$ simultaneously satisfies: the cusp class at each puncture; the splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (exponents $\sum_k A_k = \mathrm{diag}(2, 1, 1)$); the *path-ordered* monodromy at infinity is trivial, $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$, i.e. $M_1M_2M_3M_4 = \mathbf{1}$; and the representation is *irreducible* — **case A** (non-trivial $SU(3)_F$ mixing, not the diagonal case B). So a valid (U_{wall}) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope:* this is one realising point (existence + case A), likely one of a residual family; the unique $\nabla_F^\star$ still needs the $\det R = 8$ selector and c_u/c_d still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary (H2). [`verification/v33_explicit_flat_bundle.py`]

H2 bridge probed ([`verification/v34_h2_bridge_attempt.py`]). The explicit per-puncture monodromies M_k (cusp class, $\prod M_k = \mathbf{1}$ to 10^{-5}) were computed; but the natural resolvent-dressed diagonal extraction ($|\mathrm{diag}\, M_k| = (0, \frac{1}{2}, \frac{1}{2})$ times the C_6 resolvent factor) does *not* reproduce the lepton amplitudes $(0.475, 1.107, 0.917)$. So the precise $\Gamma_{ij}^{\min}$ geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family $\rho^\star$) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for $55/117$. *Resolved since (v118): the correct extraction is the resolvent determinant of the* $W(A_3)$ *monodromy, not the diagonal dressing — the probe's failure is thereby explained, and the dictionary is exact.*

Theorem U — the four-way split (sharper than one monster gate) [E]/[O]

D_4 fixes the admissible chamber, not the physical point (the D_4-fixed locus is positive-dimensional, U3); the discrete selectors $\det R = 8$, $\mathrm{SNF}(R) = (1, 1, 8)$, $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$ do the cutting. The gate splits into four pieces, three of them essentially closed:

- U_{unitary} [**E**]: parabolic degree 0 + polystable $\Rightarrow$ unitary (Mehta–Seshadri) $\Rightarrow$ $\Phi = 0$; the harmonic metric is the *finite* invariant form H ($M_k^\dagger H M_k = H$), not a Hitchin PDE ([`verification/v40_harmonic_metric.py`]).
- U_{H2} [**E**]: R, Q, K are read off the bundle (splitting type $= a$, $\det R = 8$, $\mathrm{Spec}(Q_+)$) ([`verification/v39_uwall_selectors.py`]).
- U_{Λ^2} [**E**]: **Readout Rigidity** — on the discrete stratum $\mathcal{S}_{8,Q} = \{\det R = 8, \mathrm{SNF}(R) = (1,1,8), \mathrm{Spec}(Q_+) = \{1,2,3\}\}$ the anchor-plane readout is *constant*, $\mathcal{C}_{ud}(\rho) = g_{\text{car}} \|\,\mathrm{Pl}_{\mathbf{1},a}(K)\|_1/(N_{\text{fam}}^2 \Delta_Q) = \frac{55}{117}$, *independent* of the continuous D_4 position — so c_u/c_d needs no point uniqueness ([`verification/v49_readout_rigidity.py`], [`verification/v42_exterior_leg.py`]).
- U_{point} [**O**]: full uniqueness of $\rho^\star$ — needed only for the *complete* $U_f^\star$ amplitude matrix, *not* for c_u/c_d.

Headline: the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; only U_{point} stays open.

The simple closure ([`verification/v71_simple_r_bridge.py`]). The selector stratum $\mathcal{S}_{8,Q}$ is now *fully derived*: $\det R = 8 = \mathrm{rank}\, E_8$ with $\mathrm{SNF}(R) = (1, 1, 8)$ is the lattice invariant (cf. $\det Q = 3 = N_{\text{fam}}$, v70), and $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$ is the D_4-equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants $(\det Q, \det K, \det R, \det L) = (3, 4, 8, 20)$, product $1920 = |W(D_5)|$. So by Readout Rigidity the quark *ratios* ($c_u/c_d = g_{\text{car}} \cdot 11/(N_{\text{fam}}^2 \cdot 13) = \frac{55}{117}$, etc.) are *pure integer Plücker readouts — no transcendental monodromy solve.* The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece, U_{point}, is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensionful scale, not a missing computation.

Gate 1 complete: $U_{\text{point}} \to v_{\text{geo}}$ **([`verification/v75_upoint_to_vgeo.py`]).** The absolute amplitudes are not free either. The lepton amplitudes are exact ($c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, v20), every cross-sector c-ratio is closed (Plücker), and each *sector product* is a clean $\varphi_0^{\rm ret}$-power (Grand Mass Volume, v46: $\det M_{\rm sector} \sim (\varphi_0^{\rm ret})^{6,9,10}$; the lepton product is $\frac{16}{7}\cdot\frac{4}{3}\cdot\frac{7}{6} = \frac{32}{9} = 2^{g_{\rm car}}/N_{\rm fam}^2$). Since (pairwise ratios) + (product) $\Leftrightarrow$ (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale* $v_{\rm geo}$. **So $U_{\rm point}$ is not an independent gate: it reduces to $v_{\rm geo}$, the *same* one irreducible dimensionful anchor as gravity's $1/G$ (v68) — the two [O] anchors collapse to one.**

Research Contract 2 — ($G_{\rm metric}$)

Goal. Construct the TFPT metric measure

$$Z[J] = \lim_{\chi\to\infty} \int_{\mathcal{G}_\chi/\mathrm{Diff}} \exp\bigl(-S_{\mathrm{rel},\chi}[g, A, \psi] + J\cdot O\bigr)\, d\mu_{\mathrm{Cal},\chi},$$

$$S_{\mathrm{rel},\chi} = \mathrm{Tr}\, f\Bigl(\tfrac{D_{\rm rel}+A_\Sigma}{\chi}\Bigr) - \mathrm{Tr}\, f\Bigl(\tfrac{D_{\rm ref}}{\chi}\Bigr) + \tfrac{i\pi}{2}\Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed $R + R^2$ equation.

Lemma 8 (G0–G1 — configuration space & BRST/Hodge Fredholm)**.** *For $s > 5/2$ a Sobolev space of Θ-symmetric metrics on the doubled seam collar M_Σ admits a gauge slice; the gauge-fixed gravitational complex $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2T^*M) \xrightarrow{K^*} \Omega^0(TM)$ with $\Delta_{\rm gh} = K^*K$ and $\Pi_{\rm phys} = \Pi_{\rm BRST/Hodge}\Pi_{TT}$ is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (*not *rhetorically).* Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).

Lemma 9 (G2 — finite relative spectral action, *heat-kernel grounded* [E])**.** *$S_{\mathrm{rel},\chi}$ is finite, differentiable, with a local Seeley–DeWitt expansion* $\mathrm{Tr}\, f(D^2/\chi^2) \sim f_4\chi^4 a_0 + f_2\chi^2 a_2 + f_0 a_4 + \cdots$. *For the Lichnerowicz Dirac operator $D^2 = \nabla^2 + \frac{R}{4}$ (so $E = -\frac{R}{4}$, spinor trace $\mathrm{tr}\,\mathbf{1} = 4$) the Gilkey coefficients give, per $(4\pi)^{-2}$,*

$$a_2 = \tfrac{1}{6}\,\mathrm{tr}(6E+R\,\mathbf{1}) = -\tfrac{R}{3} \quad (\textit{Einstein–Hilbert } \textstyle\int R), \qquad a_4\big|_{R^2} = \tfrac{1}{360}\,\mathrm{tr}(180E^2+60RE+5R^2\mathbf{1}) = \tfrac{R^2}{72}$$

(the R^2/Starobinsky term), so the spectral action structurally *produces* $S_{\rm eff} = \int\sqrt{g}\,\frac{\bar{M}_{\rm Pl}^2}{2}(R+\frac{R^2}{6M^2})+\cdots$ *with the scalaron mass ratio fixed by the cutoff moment, $M^2/\bar{M}_{\rm Pl}^2 = 6(4\pi)^2/f_0$. The TFPT closure $M^2/\bar{M}_{\rm Pl}^2 = c_3^7$ fixes $f_0 = 6(4\pi)^2/c_3^7$, i.e. $M = c_3^{7/2}\bar{M}_{\rm Pl} = 3.06\times10^{13}$ GeV — the boundary normalisation c_3* is *the gravitational scale — and hence the Starobinsky slow-roll forms $n_s \simeq 1 - \frac{2}{N}$, $r \simeq \frac{12}{N^2}$. The $R + R^2$* structure *is convention-independent (standard Gilkey); the precise rational $\frac{1}{72}$ and factor 6 are the standard Dirac conventions used here ([verification/v36_spectral_action_g2.py], [verification/v28_gravity_fR.py]).*

The simpler resolution: "full QG" for physics is the gap-decoupled admissible sector ([E]). The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5): $2\|V_{\rm metric}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a *strictly positive* effective gap $\Delta_{\rm eff} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

$$\boxed{\begin{array}{c}\text{admissible IR sector} = \text{closed under stated RP;}\\ \text{ambient metric measure (G6)} = G_{\rm metric}\text{ gate}\end{array}}$$

G2 supplies the local action ($R+R^2$, heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce ($G_{\rm metric}$) to the ambient projective limit (G6). **Precise status (Alessandro's**

boundary): the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure remains a *strict TOE completion target*. Its absence does *not* affect the bounded IR claim, but it *does block certification as a strict physical TOE* — so we do not call it "mere mathematical completeness". [`verification/v36_spectral_action_g2.py`]

The glue Q-system: the index-4 object of the G_{net} statement exists explicitly ([E]/[C]). The G_{net} target — *"the seam–Calderón inclusion has Jones index $4 = |\mu_4|$"* — no longer refers to an unknown object. On the glue group algebra $A = \mathbb{C}[\mathbb{Z}_4]$ (multiplication $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$, unit δ_0) *all* Longo Q-system axioms hold exactly [E]: associativity, unit law, the Frobenius identity $(m\otimes 1)(1\otimes m^*) = m^*m = (1\otimes m)(m^*\otimes 1)$, and specialness $mm^* = |\mathbb{Z}_4|\,\mathbf{1}$ — a special C* Frobenius algebra with

$$\boxed{\dim\theta = 4 = |\mu_4| = [B:A]}$$

exactly the Carrier Index Lemma value (KLM: $\mu_{\text{carrier}} = 16 = 4^2\mu_{E_8}$ re-verified). *Locality = isotropy [E]:* on the Lagrangian glue $\langle(1,1)\rangle$ the discriminant form gives $q(k(1,1)) = k^2 \in \mathbb{Z}$ for every k — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step [E]:* $\{0,2\} < \mathbb{Z}_4$ closes and gives the unique intermediate Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ tower step, with index factorisation $4 = 2\times 2$. *R2 sharpened [C] (recorded, not claimed):* what remains is the operator-algebraic *identification* "the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system" — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

The hull is the graded module of the Q-system ([`verification/v128_graded_hull.py`]**).** The Q-system is not merely an abstract companion of E_8 — E_8 *realises* it [E]: the 240 roots, built as explicit norm-2 vectors in $D_5 \oplus A_3$+glue coordinates, decompose as $240 = 52+64+60+64$ over the glue $\mathbb{Z}_4$ (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root, $\operatorname{coset}(r_1{+}r_2) = \operatorname{coset}(r_1) + \operatorname{coset}(r_2) \bmod 4$:

$$\boxed{E_8 \text{ is a } \mathbb{Z}_4\text{-graded Lie algebra over the carrier; the grading group is the glue = the Q-system}}$$

— the checksum hull is the Q-system's graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that E_8 already *carries*. [`verification/v125_glue_qsystem.py`]

Graded Frobenius ([`verification/v143_graded_frobenius.py`]**).** The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside E_8 [E]: (i) *coset duality* — $\operatorname{coset}(-r) = -\operatorname{coset}(r) \bmod 4$ for all 240 roots ($-C_1 = C_3$ bijectively, C_0, C_2 self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing $g_k \times g_{-k} \to \mathbb{C}$; (ii) the glue character average $\frac{1}{4}\sum_k i^{k\deg}$ is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index $|\mathbb{Z}_4| = 4$); (iii) each glue sector is *one* $W(D_5)\times W(A_3)$ orbit with multiplicity-one dimensions $64 = 16\cdot 4 = (\mathbf{16},\bar{\mathbf{4}})$, $60 = 10\cdot 6 = (\mathbf{10},\mathbf{6})$ — simple currents, sector ring $= \mathbb{Z}_4$ = the $\mathbb{C}[\mathbb{Z}_4]$ fusion, halfway subgroup $\{0,2\} = SO(16)_1$. *R2 residue [C] (final form):* exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

Where that statement lives: the NS/R sector census ([`verification/v148_fock_census.py`]**).** The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* $(D_5, SO(6))$ weights, and integer weights lie in the even discriminant classes — so the NS module supports only $\{(0,0),(2,0),(0,2),(2,2)\}$: **the odd glue classes $k(1,1)$ — the actual μ_4 simple currents — have zero NS support; they are twisted (Ramond-type) modules.** The Ramond zero-mode module ($2^8 = 256$ ground states) carries exactly the four odd class pairs, 64 each, and splits $128+128$ into the odd sectors of the *two* Lagrangian glues $\langle(1,1)\rangle$ and $\langle(1,3)\rangle$ (v92): *choosing the glue is choosing the Ramond projection* (128 = one $SO(16)$ spinor chirality), and E_8 assembles as $248 = 120_{\text{NS}} + 128_{\text{R}}$. *Consequence [C]:* the index-4 extension cannot even be *stated* on the untwisted module alone; R2's one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census).

Lemma 10 (G3–G4 — reflection positivity, fixed and integrated)**.** *For each Θ-invariant g in the Calderón ball the physical boundary kernel $\mathcal{K}_g = P_{\rm adm}\Pi_{\rm phys}\mathcal{N}_g^{TT}\Pi_{\rm phys}P_{\rm adm}$ is reflection positive, $\langle\Theta F, F\rangle_{\mathcal{K}_g} \geq 0$ (G3); and $d\mu_{{\rm Cal},\chi}$ is Θ-invariant, supported where G3 holds uniformly, so $\int\langle\Theta F, F\rangle_{\mathcal{K}_g}\, d\mu_{{\rm Cal},\chi} \geq 0$ (G4). **G4 is the decisive step:** RP after integrating over dynamical metrics, not merely at fixed background.*

Lemma 11 (G5 — gap dominance [E])**.** $2\|V_{\rm metric}\|_{\rm rel} < \Delta = 6\log\frac{3}{2}$, *with* $\|V_{\rm metric}\|_{\rm rel} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$, *hence* $2\|V\| = 0.7852 < 2.4328$ *and* $\Delta_{\rm eff} \geq \Delta - 2\|V\| > 1.647 > 0$. *The numerical inequality is trivial ([`verification/v29_research_contract_certs.py`]); the work is the operator-norm bound (hard).*

Sugawara reading + atomic OS moment ([`verification/v171_os_moment_cluster.py`]). The G5 budget is the E_8 Sugawara constant: $248c_3^2 = \frac{31}{8\pi^2}$ with $31 = 1 + h^\vee(E_8)$ and $c((E_8)_1) = 248/31 = 8$, so gap dominance reads $|R^+(A_3)|\log(N_{\rm fam}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$ — the A_3 transfer gap beats the E_8 Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*: ${\rm spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ makes every two-point correlator $C(n) = A + B(64/729)^n + C(1/729)^n$ the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant 729/665, correlation length $\xi = 1/(6\log\frac{3}{2}) = 0.4111$).

Theorem 1 (Decoupling theorem — physical low-energy claims do *not* need the full $G_{\rm metric}$)**.** If *the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,* $2\|V_{\rm metric}\| < \Delta = 6\log\frac{3}{2}$ *(G5),* then *the admissible IR measure is stable under the ambient completion: a strictly positive effective gap $\Delta_{\rm eff} = \Delta - 2\|V\| > 0$ protects the admissible sector from the (un-built) ambient/deep-UV, so every* physical low-energy *read-out (masses, mixings, α^{-1}, the $R{+}R^2$ regime) is independent of the open ambient measure. Symbolically*

$$\boxed{\mathrm{RP}\ +\ 2\|V\| < \Delta \implies \textit{admissible IR measure stable under ambient completion}\ .}$$

Thus ($G_{\rm metric}$) *is* not *a diffuse "full quantum gravity missing" block: it is the sharply delimited* ambient projective limit *(G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on. **Typing:** the inequality is machine-checked [E]; the operator-norm bound on $\|V_{\rm metric}\|$ is the conditional input.*

Lemma 12 (G6–G7 — projective limit & Ward identities)**.** *A projective system μ_{χ_n} with $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$, uniform moment bounds $\sup_n\int\|g\|^p_{H^s}\,d\mu_{\chi_n} < \infty$, tightness and regulator-independence yields $\mu_{\rm QG} = \lim\mu_\chi$ on $\mathcal{G}/{\rm Diff}$ (G6, the hardest step); the limit obeys the BRST Ward identities $\langle s\mathcal{O}\rangle = 0$, whence $\nabla^\mu T_{\mu\nu} = 0$ and the η boundary phase cancels the APS anomalies (G7).*

Gate 2 reduction: G6 is a *boundary* projective limit, not a bulk one ([`verification/v76_gmetric_reduction.py`])

The exact decoupling margin is $\Delta_{\rm eff} = \Delta - 2\|V\| = 6\log\frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, with $\|V_{\rm metric}\|_{\rm rel} \leq 248\,c_3^2 = \dim(E_8)\,c_3^2 = \frac{31}{8\pi^2}$ ($31 = 2^{g_{\rm car}} - 1$). **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so *G6* is a *boundary* projective limit. The conditional theorem is then

$$\boxed{\mathrm{RP}(\text{seam-Calderón boundary kernel}) + \text{tightness} \implies G_{\rm metric}\ \text{closes}}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — reduced, not closed. So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) reduced to a boundary measure, no longer a diffuse "full QG missing". [C]/[O]

The classical field equation, parameter-free: entanglement equilibrium gives the linearised Einstein equation with G from c_3 ([`verification/v358_grav_entropy_equilibrium.py`])

While Gate 2 (above) reduces the metric-sector *measure*, the classical metric-sector *field equation* is supplied directly by the entanglement first law $\delta S = \delta\langle K\rangle$ (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms. **Parameter-free coefficient [E].** Equilibrium gives $G_{ab} = c_3^{-1}T_{ab}$ with $c_3^{-1} = 8\pi$ *fixed*: all four c_3-roles are mutually consistent — the Unruh factor $1/(2\pi) = 4c_3$, the EH action $1/(16\pi) = c_3/2$, the Einstein coefficient $8\pi = 1/c_3$, the Bekenstein density $1/4 = 1/|\mu_4|$ — so *no free dimensionless Newton dial* enters. **Thermo = geo coincidence [E] (new).** The first law requires the coefficient $2\pi/\eta$ (η the entropy density); the geometry gives $1/c_3 = |\mathbb{Z}_2|\,2\pi\,\chi(S^2)$ (one-sided Gauss–Bonnet, v58); these are the *same* equation iff $|\mu_4| = |\mathbb{Z}_2|\,\chi(S^2) = 4$ — a TFPT atom identity (v216). So the thermodynamic (first-law) and geometric (Gauss–Bonnet) origins of c_3 are *one*. **The matter side, assembled [C] (closing J3).** The ball modular Hamiltonian is the boost (Casini–Huerta–Myers; geometric by Bisognano–Wichmann, v323), $K_B = 2\pi\int_B w\,T_{00}$, $w = (R^2 - r^2)/(2R)$; the 3-ball weight integral is $\int_B w\,d^3x = 4\pi R^4/15$, so $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\,\delta\langle T_{00}\rangle$ — the concrete matter boost flux of TFPT's stress tensor. **The entropy density, atom-fixed [E].** The Bekenstein numerator $1/4 = 1/|\mu_4|$ (v57) and the entanglement normalisation = the central charge $c = g_{\rm car}+N_{\rm fam} = 8$ pin the dimensionless coefficient; the area-law form is established (v59). **Residual [O]:** only (i) the linear→full non-linear covariant extension and (ii) the absolute scale $v_{\rm geo}$ (the Planck-area unit) remain — *no free dimensionless parameter survives in the gravity coupling.*

And that boundary measure is a *rigorously constructed* net ([`verification/v77_e8_conformal_net.py`]). The level-1 central charges $c = \dim\mathfrak{g}/(1 + h^\vee)$ give $c(E_8)_1 = \frac{248}{31} = 8 = \operatorname{rank} E_8$, $c(D_5)_1 = \frac{45}{9} = 5$, $c(A_3)_1 = \frac{15}{5} = 3$, and $c(D_5)+c(A_3) = 5+3 = 8 = c(E_8)$ is a *conformal embedding* $(D_5)_1\times(A_3)_1 \subset (E_8)_1$ (coset $c = 0$). Now $(E_8)_1$ is the holomorphic $c = 8$ *E_8-lattice* VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** *identify the seam-Calderón boundary measure with the $(E_8)_1$ lattice net* (carrier = $(D_5)_1 \times (A_3)_1$ subnet), and the gap $\Delta_{\rm eff} > 0$ supplies the clustering/tightness — so $G6$ is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

The Simple-Current Extension Theorem — the central $G_{\rm net}$ closing statement [E]/[C] ([`verification/v154_simple_current_theorem.py`])

The conditional theorem can now be stated as one clean *simple-current extension.* Let $A = (D_5)_1 \otimes (A_3)_1$ be the carrier net and $L = \langle(1,1)\rangle \subset \mathrm{Disc}(D_5) \oplus \mathrm{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$ the isotropic μ_4 glue line ($q(k(1,1)) = k^2 \in \mathbb{Z}$). Then the local extension $B = A \rtimes L$ has

$$\boxed{[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5{+}3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1}$$

so B is *holomorphic* (KLM μ-additivity), and a holomorphic c=8 chiral net is the lattice net of the *unique* even-unimodular rank-8 lattice — hence $\boxed{B \cong (E_8)_1}$ (the same-c rival $SO(16)_1$, μ=4, is excluded). Every algebraic ingredient is machine-checked: the $\mathbb{C}[\mathbb{Z}_4]$ Longo Q-system ([`verification/v125_glue_qsystem.py`]), the $\mathbb{Z}_4$-graded Frobenius E_8 hull ([`verification/v143_graded_frobenius.py`]), the Ramond glue sectors ([`verification/v148_fock_census.py`]). **Consequence:** $G_{\rm net}$ is internally *closed* if the seam–Calderón boundary net is *defined* to be A; it stays the *one* external identification "the RP seam–Calderón net is A" if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not "find a quantum gravity".

And that one identification reduces to a single shared premise ([`verification/v155_quasifree_boundary.py`]). The identification splits (v110) into (a) "the seam Calderón involution is sheet-odd" — the double-cover deck, the same $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2\cdot 4\pi}$ (structural) — and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the bulk covariance, and the compression of a contraction ($i\Gamma$ with spectrum in $[-1, 1]$) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a

Pfaffian of Γ_d). Hence (b) *follows* from "the seam bulk is a reflection-positive free field" — the *same* premise already load-bearing in the area law ([verification/v59_area_law_evidence.py]) and the EH mechanism ([verification/v150_replica_eh_model.py]/[verification/v151_bfk_split.py]). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by G_{net}, the area law and R3, and given it the chain Gaussian bulk ⇒ quasi-free boundary ⇒ one kernel = net (v113) ⇒ carrier $A \Rightarrow (E_8)_1$ is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

The construction, exhibited and verified ([verification/v156_seam_net_construction.py]). The "$B \cong (E_8)_1$" step is no longer an assertion — the chiral net is built explicitly and the identification *checked*. The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode e^{ikx} by the harmonic extension $e^{ikx-|k|y}$: $-\partial_n u|_0 = |k|\,u$, so for a free bulk the seam operator is the massless chiral dispersion $\omega_k = |k|$ that defines the free-fermion net on S^1. Sixteen Majoranas give $c = 8 = 5{+}3$; the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ (the $SO(16)$ bilinears plus the Ramond spinor, Frenkel–Kac); and the holomorphic character is

$$\boxed{\chi = \frac{E_4}{\eta^8} = q^{-1/3}\big(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \cdots\big) = j^{1/3}}$$

(verified by q-series), the 248 at level 1 being $\dim E_8$ — so the net *is* $(E_8)_1$. *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi–Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate ([verification/v157_rigid_fixed_point.py]). Three facts click together. (i) The DtN/Calderón symbol is *universally* $|k|$: the boundary operator of *any* second-order elliptic bulk is an order-1 ΨDO with principal symbol $|\xi|_g$ (Lee–Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic $c{=}8$ theory is *rigid*: every field has weight $(h, \bar h{=}0)$, an exactly-marginal deformation needs $(1,1)$, and with $\bar h{=}0$ *no* $(1,1)$ field exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same* $|\mathbb{Z}_2|$ that halves the one-sided S^2 Gauss–Bonnet ($c_3 = \frac{1}{|\mathbb{Z}_2|\,2\pi\,\chi(S^2)} = \frac{1}{8\pi}$) labels the sheet *and* is the Ramond/GSO projection giving $\mu{=}1$ ($248{=}120_{\text{NS}}{+}128_{\text{R}}$) — one object, three roles, so the geometry fixing c_3 is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from "prove the seam is Gaussian" to "the gapped ($\Delta{=}6\log\frac{3}{2}{>}0$) boundary flow reaches the holomorphic fixed point" — given which, freeness is *forced* by rigidity + uniqueness ($(E_8)_1$, v83), not fine-tuned. A strictly smaller and simpler residual.

And the destination is a *proven* stable fixed point ([verification/v158_fixed_point_stable.py]). An exact operator-dimension count settles it: the chiral Majorana has $h{=}\frac{1}{2}$, the bilinear current $h{=}1$, the quartic $h{=}2$. The relevant window $0 < h < 1$ for bosonic deformations is *empty* (bosonic chiral operators have integer h, lowest 1) — nothing relevant flows the free theory away; the $h{=}1$ currents are chiral (v157), not $(1,1)$-marginal; and the lowest genuinely-interacting operator (the quartic, $h{=}2{>}1$) is *irrelevant*, as are all higher $2k$-fermi operators ($h{=}k$). So the free chiral $c{=}8$ fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ($\Delta{=}6\log\frac{3}{2}$) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise ([verification/v160_seam_gaussianity_from_pf.py]). Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants $\kappa_{2n}{=}0$ (the signed set-partition/Pfaffian inversion of $\langle\psi_1\cdots\psi_{2n}\rangle = \mathrm{Pf}(\Gamma)$ leaves only the degree-2 kernel — the v113 "one kernel = the net" property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if κ_{2n} were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap > 0 suffices; the value $(2/3)^6$ only sets the rate). *The honest residual:* v56's gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries $\binom{16}{4}{=}1820$ and $\binom{16}{6}{=}8008$ higher-cumulant directions — the Schwinger cone the transport must contract has dimension $\gg 3$, and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is**

equivalent to one strictly smaller, internal, exact-testable statement: the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

And that full-cone premise is now *discharged* to a single finite one-particle statement ([`verification/v161_one_particle_reduction.py`]). The lever is Bogoliubov second quantisation: a quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dimensional Majorana space H_Σ, and the Fock space grades by correlator degree, so $\Gamma(t)$ acts on the degree-m sector as the exterior power $\Lambda^m(t)$. Three exact consequences follow (all machine-checked): (i) $\mathrm{spec}(\Gamma(t)|_{\deg m})$ is the set of products of m one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap *equals* the one-particle gap**: the sub-leading eigenvalue of $\Gamma(t)$ over *every* degree is $(2/3)^6$ (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is Λ^* of the fixed subspace — the disconnected Wick powers of K_Σ — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation $\Leftrightarrow$ one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the $120 = \dim \mathfrak{so}(16)$ weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport's one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8 K_Σ polarisation (v113), sub-leading gap = $(2/3)^6$ (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of (A_2), now with the infinite-dimensional cone eliminated [E]/[O].

And that finite statement carries no *new* open content: it factors into the two already-open items ([`verification/v162_seam_transport_identification.py`]). Take the one-particle symbol apart into its two data, the gap value (β) and the fixed space (α). (β) *is forced*: a μ_4-equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection, $\sum_{k=0}^{3} U^k X U^{-k} = |\mu_4|\,\mathrm{diag}(X)$ for $U = \mathrm{diag}(1, i, -i)$, and the commutant of U is exactly the diagonal algebra — so its spectrum is the cusp weights $\mathrm{spec}(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (the parabolic residue, v115), and the transfer eigenvalue $(1-\alpha)^{p_2}$ with p_2=6=$|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of the ω_k=$|k|$ free dispersion (v113/v156), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is* μ_4-equivariant and admissible — which is the R1 "seam clock" identification (the transfer operator as the seam's gapped generator) — together with the A_2 net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into $A_2 + R_1$ with every number forced. The "free-bulk cloud" is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) [E]/[O].

And both residual identifications are pinned to their floor, so (A) has no independent open gate left ([`verification/v165_premise_a_closure.py`]). A_2 (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic c=8 lattice net *is* $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally $|k|$ (v156/v157). So A_2 re-types [O] → [C] (closed by assembly). R_1 (the seam clock) reduces to `GATE.QGEO`: a μ_4-equivariant flat gapped generator on the carrier is *unique* (= $A_0^\star$, v115/v116), so its gap $(2/3)^6$ is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the μ_4-punctured sphere $\mathbb{P}^1 \setminus \mu_4$ — whose cohomological half is already established (v137, b_1=3=N_{fam}). That is `GATE.QGEO`, an already-open realisation premise, *not* a new gate. By the No-Unit Theorem (v153) the irreducibles stay exactly $\{\pi, v_{\mathrm{geo}}\}$ — closing (A) adds none. **Final accounting:** premise (A) = (A_2 assembly, [C]) + (R_1=`GATE.QGEO`, already open) + the $\{\pi, v_{\mathrm{geo}}\}$ irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate [E]/[O].

And the two structural residuals — net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the one irreducible open premise ([`verification/v175_net_existence_full_cone.py`]). The CAR second-quantisation functor $\Gamma(t)$=$\bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the *complete* Fock space $\Lambda^\bullet(\mathbb{R}^{16})$

(dim 2^{16}=65536); by the exterior-power spectrum theorem every eigenvalue of $\Gamma(t)$ is a subset product of spec(t), so checking all 2^{16} products shows $\Gamma(t)$ is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity 2^8=256 (the wedges of the rank-8 K_Σ polarisation) and sub-leading eigenvalue exactly the one-particle gap $(2/3)^6$ — so full-cone RP/gap reduce *for every* m (not only $m \le 6$, v161) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem.* And A_2 is now an *assembled and verified* certificate: the $(E_8)_1$ character E_4/η^8=1+$248q$+$4124q^2$+$34752q^3$+$213126q^4$+$1057504q^5$ (level-1=248), the embedding 248=120+128, and the E_8 Cartan matrix even with det 1 (the unique rank-8 even unimodular lattice, v83) — the net *exists* by cited theorems. **Both residuals then collapse to the *same* input**: that the actual seam Calderón kernel is this gapped one-particle contraction on $\mathbb{P}^1 \setminus \mu_4$, i.e. `QGEO.REALIZE.01`, whose finite half is proven (v168). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]— the single irreducible structural premise; net existence and full-cone RP are [E].

The No-Unit Theorem — $v_{\rm geo}$ is an irreducible metrology primitive [E]/[O] ([`verification/v153_no_unit_theorem.py`])

$v_{\rm geo}$ is not an open gap. Every load-bearing compiler datum — $g_{\rm car}, |\mu_4|, N_{\rm fam}, \operatorname{rank} E_8, \alpha^{-1}$, the determinant ladder $(3, 4, 8, 20)$, $c_3 = \frac{1}{8\pi}$, $\varphi_0^{\rm ret}$ — is mass-dimension 0 and *invariant* under a unit rescaling $L \mapsto \lambda L$, whereas a mass / energy / $1/G$ carries nonzero dimension. A map from dimension-0 inputs to a dimension-$d{\neq}0$ output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** — $v_{\rm geo}$ is a *forced input.* The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\rm point} \sim v_{\rm geo}, \qquad \tfrac{1}{G} \sim v_{\rm geo}^2, \qquad m/\mu = e^{3/4} \text{ (a pure ratio)},$$

so the irreducibles stay exactly {one dimensionful unit, π}. *Re-typing:* $v_{\rm geo}$ is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

Theorem G — the closure statement [O] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure $\mu_{\rm QG}$. Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT $R + R^2$ seam action

$$f_R R_{\mu\nu} - \tfrac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\rm Pl}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: open; the deepest item. G5 certified; the regime equation closed [`verification/v28_gravity_fR.py`].*

Certifiability and recommended order

Building block	**Machine**	**Hardness**
$W_{\rm wall}$ enumeration ($C_U^{(1)}$)	Lean / Python / Wolfram	easy (*done*)
D_4 fixed-locus polynomial ($C_U^{(2)}$)	Sage / Mathematica	medium
selector uniqueness ($C_U^{(3)}$)	interval arithmetic	medium
SNF, determinants, spectra	Lean	easy–medium
principal-symbol ellipticity (G1)	xAct / Cadabra / Lean	medium
gap inequality (G5)	Lean	easy (*done*)
operator-norm bound (G5)	analysis + numerics	hard
projective limit (G6)	not primarily machine	very hard
Ward identities (G7)	BV/BRST formal + checks	hard

Recommended order.

$$\boxed{\text{Selector-triangle pairings} \;\rightarrow\; v_{\rm geo} \text{ scale anchor} \;\rightarrow\; G_{\rm net} \text{ inclusion theorem}}$$

The flavor interface (historically $U_{\rm wall}$) is now reduced to the *selector-triangle pairings* (the dual normal pair (d, n) forcing R columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The $v_{\rm geo}$ scale anchor is the one dimensionful input (same nature as $1/G$). $G_{\rm net}$ — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit. $F_{\rm transfer}$ (source→pole / relic / cosmology) is the remaining downstream interface.

$F_{\rm transfer}$ is a typed functor, not a bag of open topics. It is standard physics fed TFPT *source* data, $F_{\rm transfer} = F_{\rm observable} \circ F_{\rm threshold} \circ F_{\rm RG}$, with exactly four interfaces:

Interface	Map	Status	Source data → observable
$F_{\rm pole}$	source→pole masses	[E] / [C]	Koide $53/54 = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra)$ is an exact operator factor [E] (the missing Sheet-Diamond corner); the pole-transfer interpretation is [C] ([`verification/v183_koide_f_corner_transfer.py`])
$F_{\rm Boltzmann}$	CP source→ η_B	[C]	$\tilde m_1 = m_3/A_\Lambda$ anchors the washout, but $M_1 = M_R\varphi_0^4$ only relocates the free input (M_R is the seesaw scale, not a compiler power), so η_B stays a *sharper scenario*, not a closure ([`verification/v184_etaB_anchored_boltzmann.py`])
$F_{\rm relic}$	$(f_a, m_a, \theta_i, N_{\rm DW}) \to \Omega_a$	[C]	the determinant-line axion can be all of DM, but only on the $\theta_i \approx 170°$ hilltop where the relic is exponentially sensitive — a viable scenario ([`verification/v185_axion_relic_solver.py`])
$F_{\rm QCD}$	$(\alpha_s, b_3, \Lambda_{\rm QCD}, \text{lattice}) \to m_p/m_e$	[O]	a QCD matching contract ($b_3 = -7$ a carrier output), *not* a compiler number — the rejected 1836 near-formula is the discipline working

A machine guard enforces the typing: the four families must always carry a [C]/[O] marker and must never be promoted to a primitive [E] compiler prediction ([`verification/v187_ftransfer_laws.py`]). Exact *algebraic* sub-parts (the Koide factor, $b_3 = -7$) may be [E]; the *physical* prediction never is.

The functor contract `CONTRACT.F.01`, four axioms ([`verification/v213_ftransfer_functor.py`]). Beyond the typing guard, the functor itself satisfies four checkable structural axioms, each instance a *discrete* compiler kernel [E] composed with an *external* continuous solver [C]: **(1) μ_4-deck equivariance** — the discrete kernels are carrier/deck invariants $\{|\mu_4|, N_{\rm fam}, g_{\rm car}, A_\Lambda\}$, and the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ *is* the μ_4-deck transfer eigenvalue (v82/v54); **(2) Plücker preservation** — the dimensionless readouts are Plücker readings ($53 = a^T(R+Q)\mathbf{1}$, $\mathrm{Pl}(F) = (1, 22, 30)$; $\|\mathrm{Pl}(K)\|_1 = 11$); **(3) positivity/stochasticity** — $\mathrm{spec}\,T = \{1, (2/3)^6, (1/3)^6\}$ is positive (RP/OS), $\kappa_f \in (0,1)$; **(4) external modules explicit** — the continuous content (pole transfer, the flavored Boltzmann network, the finite-T $\chi(T)$ ODE, the lattice C_N) is named and external, never a compiler number. So `CONTRACT.F.01` is the third research contract alongside ($U_{\rm wall}$) and ($G_{\rm metric}$): a typed functor, *not* a bag of open topics — but a consolidation, *not* a closure.

The discrete→dynamic lens: `F_transfer` is the readout end of the one principle ([`verification/v303_ftransfer_dynamics.py`], `FR.DYNAMICS.01`). The typing above is sharpened by a *dynamical* reading. The main-branch mechanism is a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update $m \leftarrow (A+2I)/4\,m \to$ the E_8 marks, gap $6\ln(3/2)$). All four `F_transfer` instances share that *same shape*: [E] **F_pole** (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F' = (2/3)^6 = \lambda_2$, the seam-clock subleading eigenvalue, so $|F'| < 1 \Rightarrow$ a unique attractor at the *seam* rate (v82); [C] **F_Boltzmann** (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$ toward the unique balance, an H-theorem); [C] **F_relic** (axion) freezes at an adiabatic fixed point (the comoving number relaxes to a constant); [E] **F_QCD** (m_p/m_e) is the 1-loop RG flow with the carrier $b_3 = -7$, whose Gaussian UV fixed point is the attractor (asymptotic freedom, $\Lambda_{\rm QCD}$ generated). *Honest boundary:* only F_pole has the seam rate $(2/3)^6$ exactly; the other three share the *shape* with external rates (thermal, cosmological, RG). So `F_transfer` is not a bolt-on but the downstream manifestation of the one discrete→dynamic principle — the compiler fixes the discrete seed + integer kernel, a gapped flow carries it to the observable — and the theory's simplicity is preserved

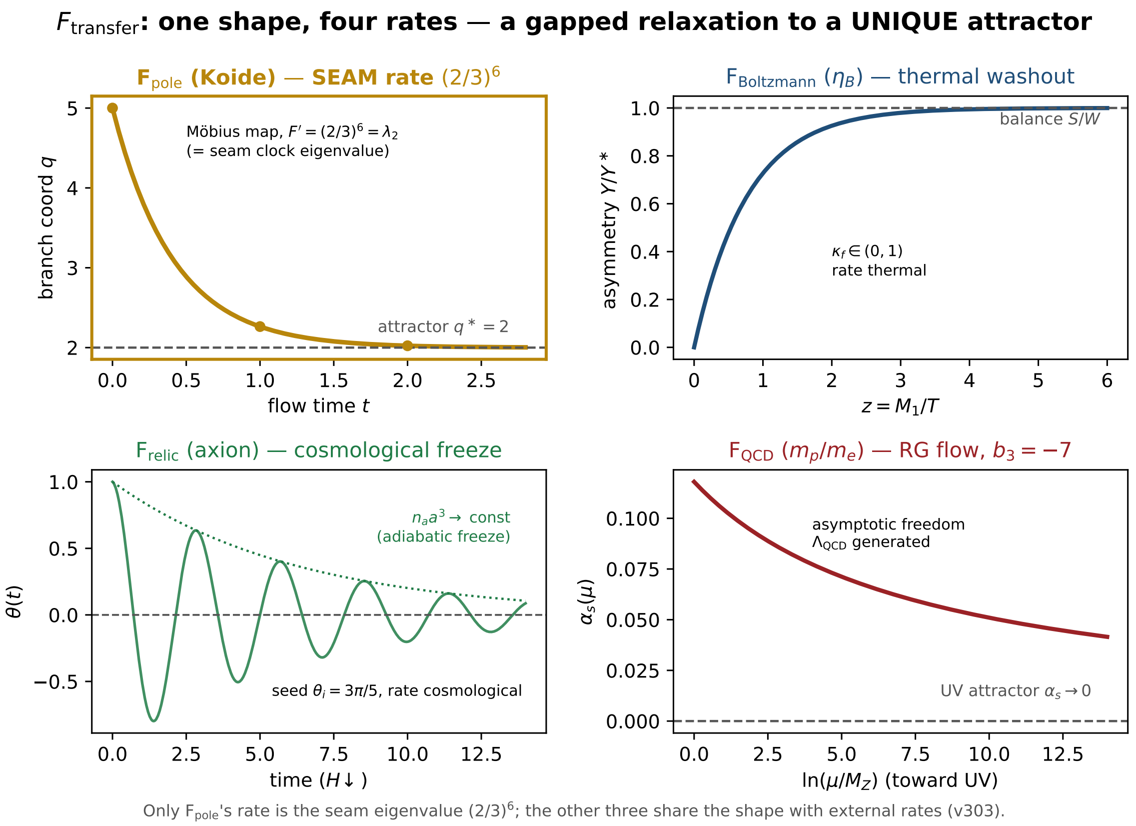

Figure 4: **The discrete→dynamic lens on `F_transfer`** (`[verification/v303_ftransfer_dynamics.py]`). All four transfer instances share *one* shape — a gapped, positivity-preserving relaxation to a *unique* attractor, the same Perron-Frobenius / H-theorem / RG-fixed-point structure as the main-branch update $m \leftarrow (A+2I)/4\,m \to$ the E_8 marks. *Only* F_{pole} (Koide, gold) runs it at the *seam* rate: its Möbius multiplier is $F'=(2/3)^6=\lambda_2$, the seam-clock subleading eigenvalue. $F_{\text{Boltzmann}}$ (η_B, thermal washout), F_{relic} (axion, cosmological freeze) and F_{QCD} (m_p/m_e, the RG flow to the Gaussian UV fixed point with the carrier $b_3=-7$) share the *shape* with *external* rates. So `F_transfer` is the readout end of the one principle, not a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam.

precisely because the external rates are honestly fenced (v187), not because they reduce to the seam. (Even F_pole's flow time is non-integer, v93: the relaxation is the *continuous* semigroup $e^{t\log F}$ generated by the discrete operator — exactly the diffusion-semigroup reading of the main branch.) So the visible residual is one geometric postulate (`QGEO.SYM.01`), one unit gauge (v_{geo}), and this one typed transfer functor — everything else is out-of-sample data.